Multi-physical analysis of the corrosion resistance of OFHC copper surfaces obtained by orthogonal cutting

I.1 Impact of machining on electrochemical behavior and corrosion resistance - literature review

As the mechanical, physical and microstructural properties of metals can be significantly affected by machining, the electrochemical properties and corrosion resistance are consequently altered. But first of all, some notions should be clarified.

I.1.1 Definitions

Metals corrosion is characterized by two steps: initiation and propagation. Initiation corresponds to the time required for physico-chemical conditions to trigger the phenomenon of corrosion. It is accomplished either by depassivation, or by exposing the non passive material by dissolving metallurgical heterogeneities. This step corresponds to the dissolution of the metal by the following oxidation reaction:

$$M \rightarrow M^{n+} + ne^{-} \qquad \textbf{Eq.1}$$

Due to this dissolution, the generated electrons are consumed by the reduction of the elements already present in the solution according to the cathodic reactions. Electrons motion imposed by these reactions is proven by the prevailing of electrical current which can be measured. The formation of oxides besides the products of corrosion can lead to re-passivation of the surface, so to the metastability of the phenomenon. For that reason, it is clear that the speed of the surface re-passivation can define the stability of the system.

I.1.1.1 Corrosion mechanisms

When two parts of a structure have different electrical potentials, they are susceptible of forming an electrochemical cell called corrosion cell. The difference in electric potential (*dep*) is either the consequence of the material heterogeneity (different phases, grains boundaries, inclusion, segregations) or of the environment (aeration, pH, convection, temperature). In the absence of electric contact between the electrodes where the difference in electric potential is observed, to each one corresponds a free corrosion potential E_{cor}. It depends of the experimental conditions and can be measured against a reference electrode.

Table 1 summarizes the origines of corrosion cells and the associates physical phenomena.

Origins of the dep	**Name of the phenomenon**
Contact between two different metals	Galvanic corrosion
Different accessibility of oxygen	Corrosion by aeration cell, crevice corrosion
Precipitation in grains boundaries	Intergranular corrosion
Local depassivation by agressive anions	Pitting corrosion
Difference in temperatures	Corrosion by thermal cell

Table 1. *Origine of corrosion cells* [1].

i) Galvanic corrosion

Galvanic corrosion is due to the formation of an electrochemical cell between two metals where the less resistant metal is degraded. It is one of the most frequent types of corrosion in aqueous environment. Zones where anodic (material corrosion) and cathodic reactions are produced (reduction of the oxidant) are different (cf. Figure 1). This localization of reactions is essentially tied to the heterogeneity of the metal, the environment and the physico-chemical conditions at the interface.

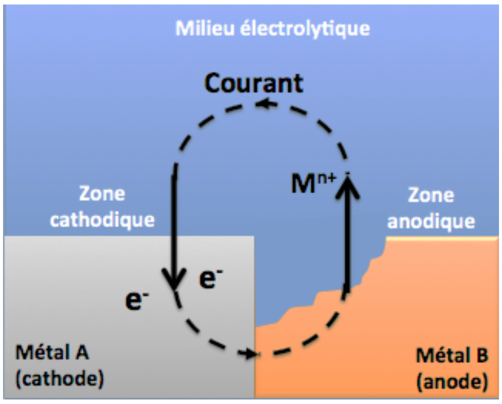


Figure 1. *Principe of galvanic corrosion.*

ii) Crevice corrosion

Crevice corrosion is associated to the presence of a confined zone with a small volume of water and a near zero flow rate (boundaries, interstices, deposits). This phenomenon induces oxygen accessibility difference or other chemical forms between two parties of one structure, creating a corrosion cell.

iii) Intergranular corrosion

Intergranular corrosion is a selective attack in grain boundaries or their immediate neighbourhood, while the remaining material is not attacked. This type of corrosion can be due to either the presence of impurities in the boundaries, or the local lack (or enrichment) of one of the alloy elements.

iv) Pitting corrosion

Pitting corrosion refers to the local attack of a passive surface. It requires the presence of agressive anions and an oxidant. It happens when the corrosion potential exceeds the pitting potential E_b. It is noticed when cavities called pitting appear at the surface. Figure 2 shows the pitting corrosion mechanism. In fact, in presence of an oxidant, a corrosion cell is produced between the exterior surface which is passive and cathodic, and inside the pitting, which is active and anodic. The ratio between the anodis/cathodis surfaces being tiny, corrosion becomes so quick inside the pitting. Cavities so created become quickly deep.

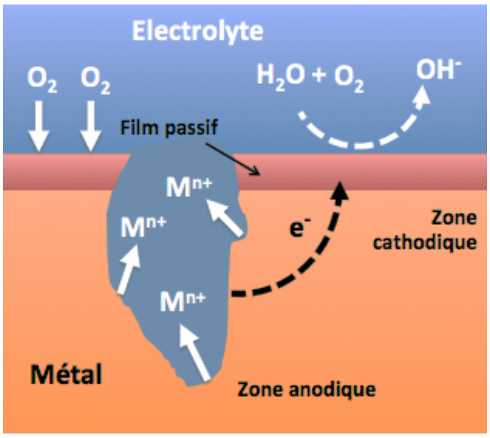


Figure 2. *Pitting corrosion: anodic and cathodic reactions.*

The value of pitting potential E_b depends on the initiating process and the cinetics of growth and repassivation, so of the chemical nature and the microstructure of the metal, of the surface state, and suddenly of the presence of inclusions, of the electrolyte chemical composition (agressive and non agressive anions), of the temperature and of the convection conditions [1].

v) Stress corrosion

Stress corrosion is the result of a common action of corrosion and mechanical stress (metal deformation under applied stresses or residual stresses). This phenomenon concerns a high number of materials, suddenly passivable ones whose protector layer is ruptured locally under the action of stresses (cf. Figure 3), leading to a localized corrosion.

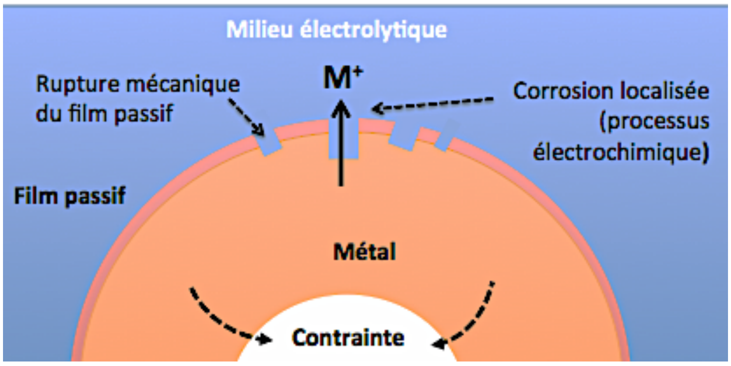


Figure 3. *Principle of stress corrosion.*

I.1.1.2 Passivity of metals

Passive metals have on their surface a thin layer of oxyde, so a passive film, which separates the metal from the electrolyte [1]. Figure 4 illustrates the variation of partial current density of OFHC copper in function of the potential obtained by a oitentio-dynamic electrochemical test.

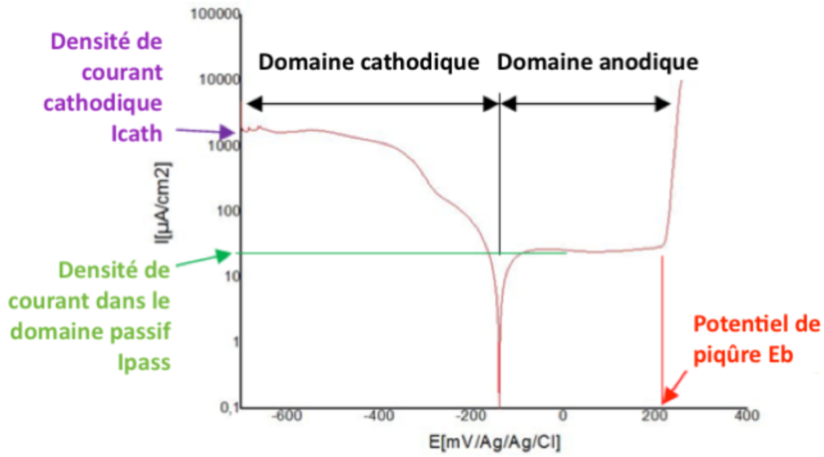


Figure 4. *Polarization curve* [2].

This kind of curves allows the measurement of the parameters necessary to describe the corrosion behavior of passivating metals. At passivation potential, the polarization curve reaches a maximum corresponding to the beginning of the anodic domain, where remains the density of current I_{pass} in the passive plate. At this potential, the metal moves from the active state to the passive one. In this domain, current density in the passive plate characterizes the dissolution rate of the passivated metal. The transpassivation potential E_b indicates the end of the passive domain. At a higher potential, the partial current density in the anodic domain raises because of the transpassive dissolution. When this dissolution is not uniforme and when pittings appear as in the case of OFHC copper in the presence of a saline solution, the transpassivation potential E_b is called the pitting potential.

I.1.2 Relation between surface integrity and local electrochemical behavior

Thanks to its functional and economical relevance, corrosion resistance of metallic materials has been the core of several investigations, especially that pitting and intergranular corrosion can present potential initiation sites of cracks, resulting into failure [3]. Among those investigations, some are interested in the relation between the parameters of surface integrity and corrosion resistance.

I.1.2.1 Metals corrosion resistance at a microscopic scale

Several metals have been the objet of study of corrosion resistance at a microscopic scale. As an example, duplex stainless steels with both austenitic and ferritic structures can be considered as metals with complex microstructure. Due to the differences of mechanical properties in both phases in addition to their morphology, a heterogeneous distribution of strains is generated [4]. This gradient of strain can affect the physico-chemical properties of the material [5]. In fact, Vignal and Kempf Vignal et Kempf (2007) have performed local electrochemical measurements using NaCl solution with the technique of the electrochemical microcell [7] on a kind of steel having a surface containing glide plans and micro-cracks. They have proposed a mecanic-electrochemical criterion aiming to enhance significantly the anodic reaction and/or the cathodic one besides reducing the pitting of the deformed material. Another study has been led by Krawiec et al. (2009) about local electrochemical behavior of the aluminum alloy AlCu4Mg1 containing large precipitates. They have demonstrated that the most active sites are also those which correspond to micro-cracks and damages of the metal matrix.

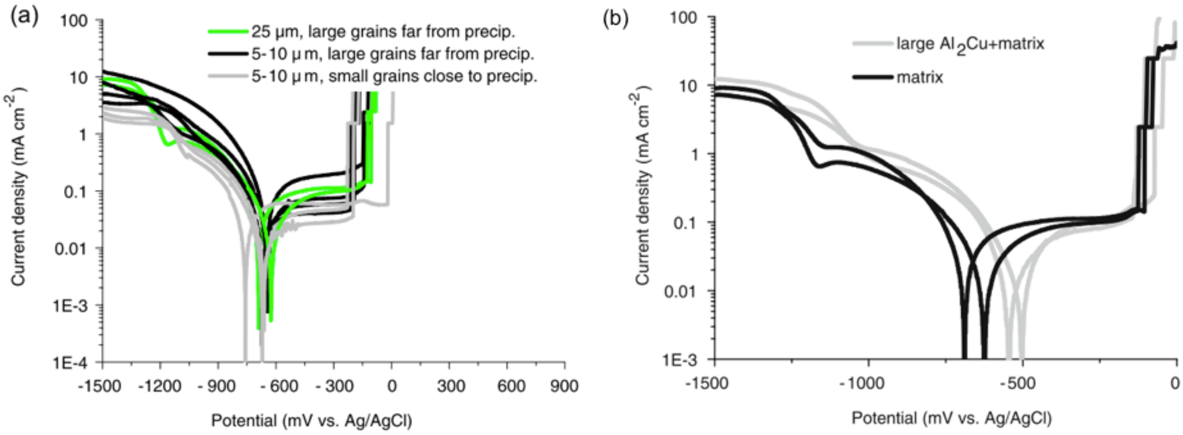


Figure 5. *Local polarization curves obtained at the scanning rate of 16.6 mV.s^{-1} on (a) specimen with $AlCu_4Mg_1$ matrix (b) specimens with Al_2Cu matrix and precipitates* [8].

However, local polarization curves performed on sites containing precipitates, but free of defaults induced by the deformations, are too close to those obtained inside the grains far from the precipitates (cf. Figure 5).

To sum up, heterogeneity of the material, damage and micro-cracks due to deformations are shown as the principle precursors of any change in the electrochemical behavior of metallic materials.

I.1.2.2 Case of copper surfaces

Having a monophasic microstructure, other aspects of surface integrity are considered while studying the local electrochemical behavior of copper. It is then necessary to quantify the residual stresses, textures, and microstructure induced by the manufacturing process. In the investigation of Bissey-Breton et al. (2007) treating the case of OFHC copper turning, the influence of the nose radius of the cutting tool, the cutting velocity and the lubricant on the residual stresses and the texture have been evaluated. Next, the effect of those surface integrity parameters on the electrochemical behavior has been studied. They have found that the stresses gradient near the surface depends on the tool geometry and the depth of cut, and that the smallest depth of cut besides the presence of a lubricant lead to a more anodic corrosion potential (cf. Figure 6). Later, Bissey-Breton et al. (2011) have found a relationship between surface integrity parameters (roughness and residual stresses) and the free corrosion potential while testing the effect of the NaCl aerated solution on local electrochemical behavior of OFHC copper.

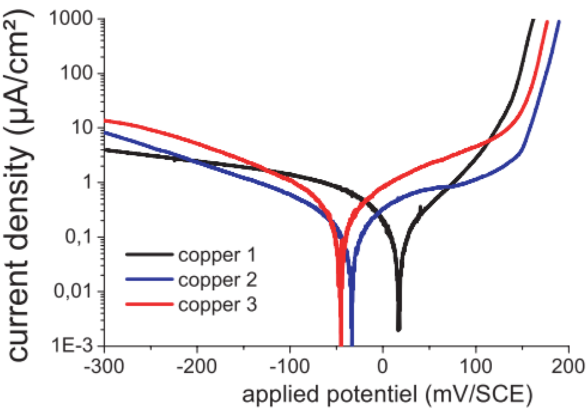


Figure 6. *Potentio-dynamic polarization curves defined on machined surfaces in 1 M $NaClO_4$ at $1 mV.s^{-1}$ for (copper 1) a depth of cut in turning a_p = 0.05 mm with lubrication (copper 2) a_p = 0.3 mm without lubrication (copper 3) a_p = 0.3 mm with lubrication* [9].

A theoretical demonstration of the impact of roughness on the mean Electron Work Function (*EWF*), in the case of copper corresponding to a reduction in *EWF*, has been performed by Li et Li (2006). Nevertheless, they have considered a raise in the fluctuation of this function. In fact, such fluctuation can promote the formation of microelectrodes and, consequently, accelerate corrosion. In addition to the microstructure and the morphology, mechanical solicitation, suddenly plastic deformation induced by the process, has an impact on the electrochemical reactivity of the surface. On the other hand, Yin and Li (2005) have found that the initial grain size, the grain boundaries and twins have an impact on the profile of subsurface residual stresses which affects corrosive wear of copper surfaces when they are exposed to a saline solution of NaCl.

I.1.3 Relationship between surface integrity and surfaces corrosion resistance in aggressive environment

Concerning machining, some studies have been performed to define the corrosion behavior of machined surfaces. In fact, Gurvich et Shubadeeva (1971) have noticed qualitatively that the different ways of surface preparation by machining have an influence on the degradation of stainless steels machined surfaces. More recent studies have been performed by Cammett and Prevey (2015) on aluminum and steel and have confirmed the influence of the exposition to a salt fog atmosphere during 500 hours on fatigue resistance. Moreover, Bouzid Sai and Triki (2006) have tested an austenito-ferritic steel inside synthetic sea water and have found that the raise in temperature induced by machining results in tensile residual stresses and then they induce a mediocre corrosion resistance comparing to that one induced by grinding or burnishing.

Concerning to OFHC copper, some works have been led on atmospheric corrosion resistance, simulated by some tests of exposition to salt fog atmosphere, suddenly of Gravier et al. (2012b) where the influence of the quadratic stress, the crystallographic texture, the lubrication and the roughness are quantified. Observations of the surfaces have revealed a general dissolution of copper under the prevailing of atacamite $Cu_2(OH)_3Cl$ and paratacamite. Then a degradation model of the machined surfaces in presence of the salt fog atmosphere has been proposed.

I.2 Experimental setup

I.2.1 Local electrochemical behavior

The object of the tests is the investigation of the effects of the cutting parameters variation on polarization curves.

I.2.1.1 Notions

It is to remind that polarization curves obtained by local electrochemical tests allow the quantification of the surface behavior in the anodic and cathodic domains. The anodic domain gives data about the material dissolution reactions.

In fact, in a given environment, the metal passivated spontaniously spontanément resists better to corrosion than an active metal. Thermomechanical and cinetic parameters controlling the spontaneous passivation of the metal should be studied. This spontaneous and stable passivation requiers both of the following conditions: ($E_p < E_{rev,ox}$) and ($i_{pass} < |i_{cath}|$) where E_p is the passivation potential, i_{pass} the density of current in the passive plate, $E_{rev,ox}$ the reversible potential of the oxidant, and i_{cath} the density of current in the cathodic domain necessary to reduce the oxidant at passivation potential. The metal which has a current density in the passive plate lower than the absolute value of the current density in the cathodic domain corresponds to a stable state of passivity, so to a small corrosion rate [1].

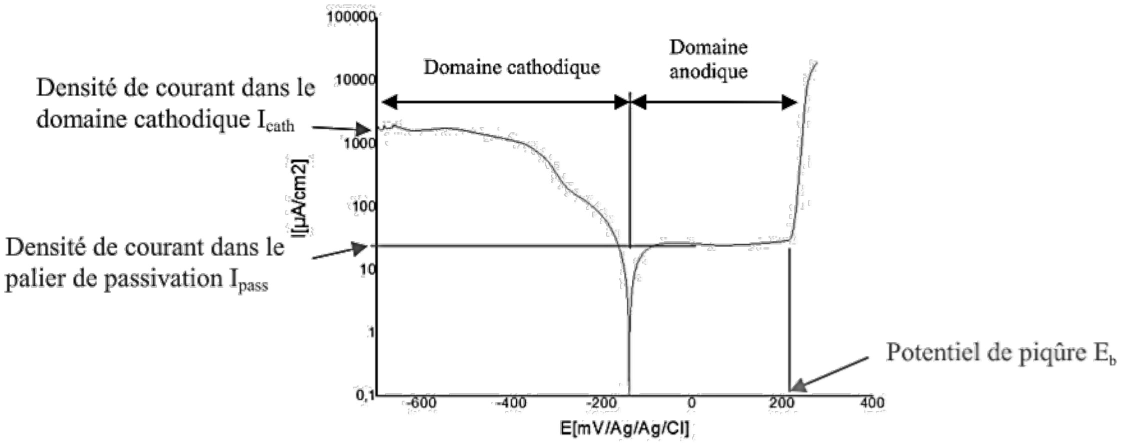


Figure 7. *Structure of a polarization curve.*

The variation of cutting parameters induces variations in the topographic, mechanical and microstructural state (parameters of surface integrity) of sublayers of the machined surface, and consequently the variation of electrochemical parameters of polarization curves. Figure 7 shows electrochemical parameters associated to those curves, which are: the current density inside the cathodic domain I_{cath}, the current density in the passive domain I_{pass}, the extent of the anodic domain, and the pitting potential E_b.

I.2.1.2 Experimental procedure

The technique of local electrochemical analysis is based on the use of a micro-capillary which confines the electrolyte at the tested surface. It requires the use of a microcell composed of a micro-capillary whose extremity is covered with silicone deposit, in order to avoid electrolyte leakage and crevasses formation during measurement. The microcell is fixed inside an optical microscope turret in order to localize with precision the microcell on the specimen's surface (cf. Figure 8). Thanks to a Faraday cage and a low noise potentiostat with high resolution equipped with double shellproof cables, all noise of electro-magnetic origin susceptible to bother the measures is deleted (cf. Figure 9).

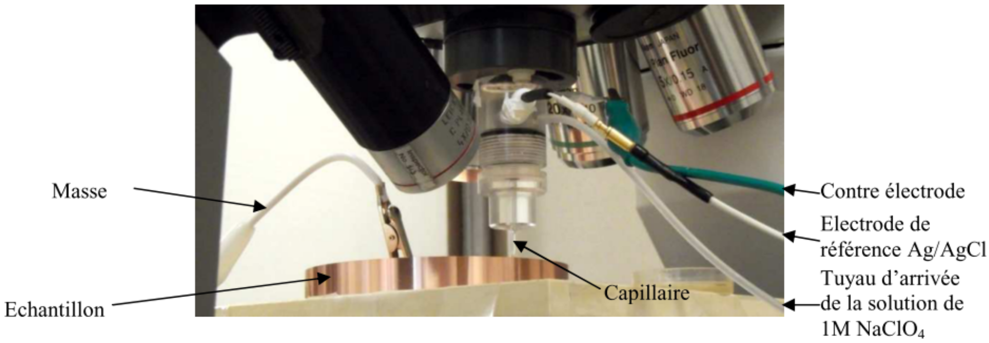


Figure 8. *Disposition of the specimen, the micro-capillary and the electrodes.*

Figure 9. *Global view of the experimental apparatus for local electrochemical tests with the microcell.*

In this study, the used electrolyte is the aerated solution $NaClO_4$ 1M. The reference electrode is Ag/AgCl. A capillary of 50 µm diameter is used. The choice of the diameter can be explained as follows: for surfaces obtained by faceturning, polarization curves concern both peaks and valleys of the machining grooves. For those measurements, the micro-capillary is positioned as in Figure 10. To be able to insert it inside the valleys, its diameter must be lower than the minimal distance between two machining grooves, which corresponds to the minimal feed f used in face turning.

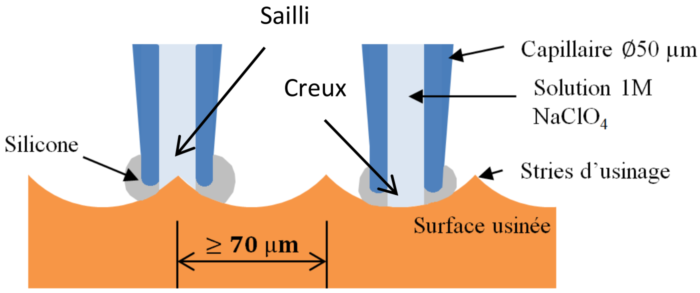


Figure 10. *Position of the microcapillary on surfaces obtained by face turning.*

I.2.2 Corrosion in salt fog atmosphere

The object of corrosion tests under salt fog atmosphere is to follow the evolution of the machined surfaces aging inside a corrosive environment.

I.2.2.1 Aging tests

The specimens of the machined OFHC copper are put into a waterproof case at a controlled atmosphere containing the salt fog 0.1 M NaCl at a continuous debit of 9 mL/h at 24 ± 1°C.

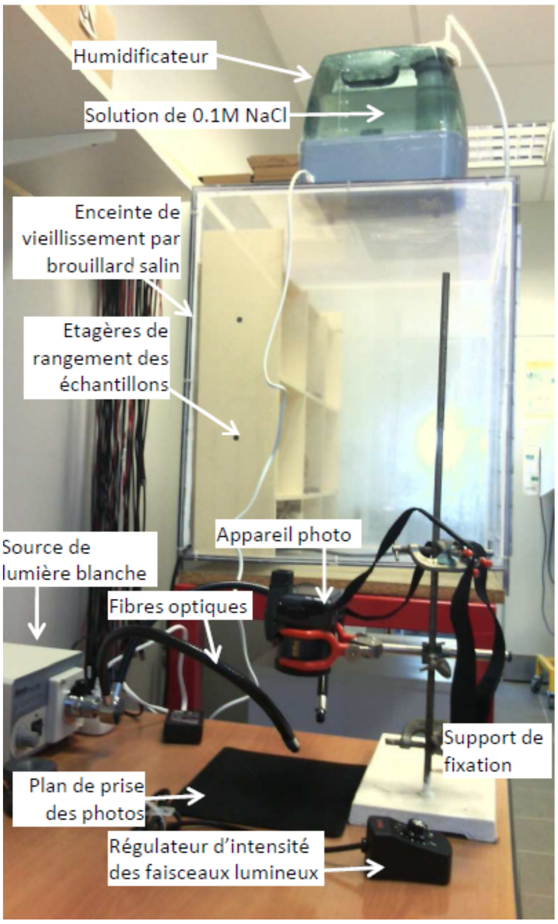


Figure 11. *Experimental setup for aging tests.*

At each specific moment of the aging time, specimens are taken in photos to calculate the evolution of the oxidized surface. The lightening with white light is controlled, and the camera is fixed in order to keep natural colors and net shape photos.

I.2.2.2 Numerical treatment

Those images are modified to isolate the studied surface from the unuseful parts of the photos (cf. Figure 12). The calculus of the percentage of green inside the surface is done with scripts developed under Matlab® environment.

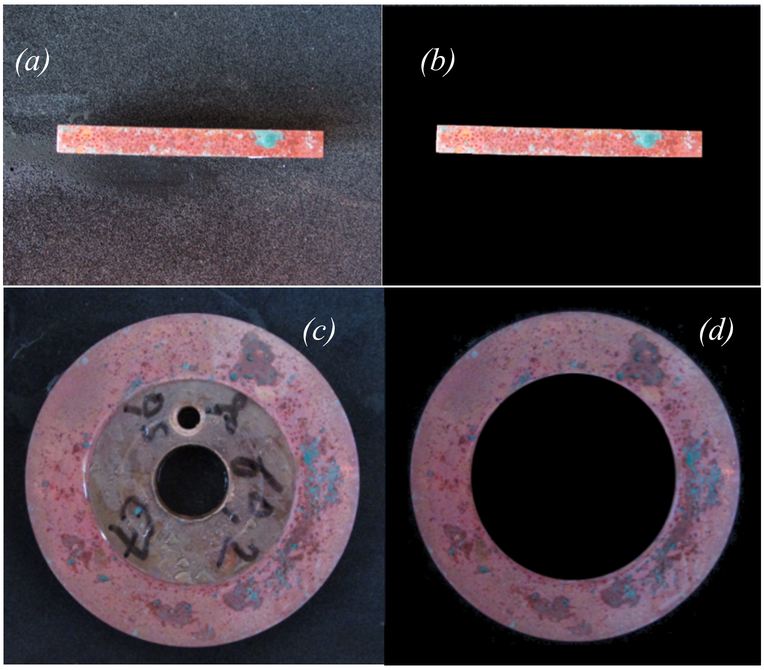
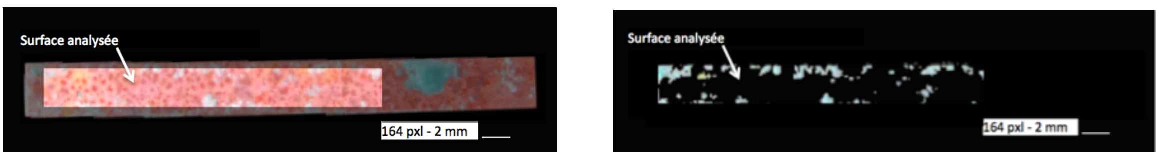


Figure 12. *Photos of the specimens' surfaces (a) obtained by planing operation (b) after photo modification (c) obtained by face turning (d) after photo modification.*

The first step performed by the script is converting the image into 256 colors. This conversion allows lighter required memory space that the image is converted into a matrix of 3 tables, where each table corresponds to a primary color (table 1 : red, table 2 : green and table 3 : blue) filled with numbers between 0 and 255 (256 shades of the color where 0 corresponds to black and 255 to white). The second step is calculating the total analyzed surface of the specimen. This surface represents only a part of the specimen where the influence of borders and cutting instabilities in the beginning of the planing operation are avoided (cf. Figure 13a). The third step is isolation of the green surface (cf. Figure 13b). The program shows the value of the green surface in pixels, and the percentage of the surface corrosion.



(a) *(b)*

Figure 13. *(a) Image converted into 256 colors (b) Isolation of the green surface.*

The procedure followed to exploit images of the surfaces obtained by face turning is similar to the previous one (cf. Figure 14).

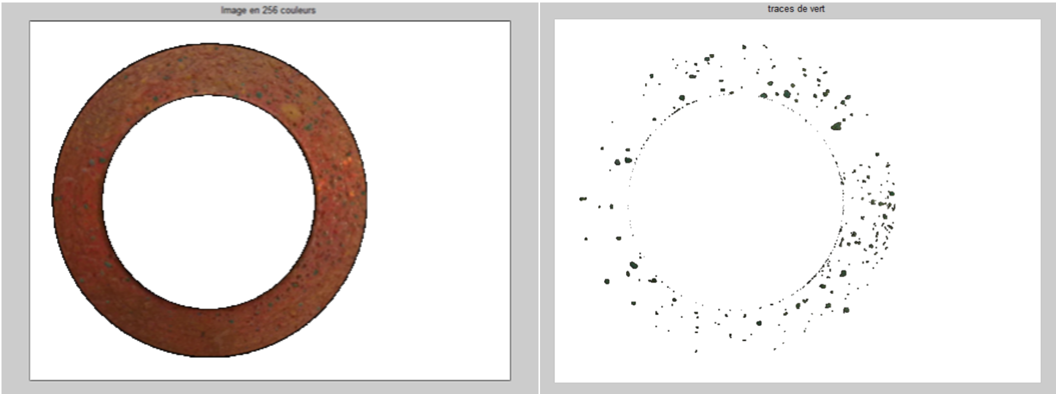


Figure 14. *Isolation des taches vertes sur les surfaces disques.*

To sum up: machined specimens of OFHC copper are put inside a controlled atmosphere composed of salt fog of 0.1 M NaCl at a debit of 9 mL/h at a temperature of 24 ± 1°C. At each measurement time, specimens are taken into high resolution photos to calculate the evolution of the oxidized surface. Then, images are treated with Matlab® software in four steps: conversion into 256 colors, filtering for the isolation of the green surface, counting of the number of green pixels, and last calculating the percentage of the oxidized surface by dividing this number by the total number of pixels representing the analyzed surface. Tests stop when the aging reaches 2556 hours.

I.2.2.3 Evaluation of the volume of copper involved in the oxidation reaction

The object of this test consists in defining the volume of copper participating into the oxidation reaction per mm^2 covered by atacamite (the green color). The followed process can be presented in five points:

1- Taking the oxidized surface in photo by a high resolution microscope (Keyence™)

2- Clean the surface from oxides and take it into photo by the microscope Keyence™.

3- Isolate the tested surface (dimensions 2 mm x 8 mm) using the microscope Alicona™.

4- Analyze the topography by optical measurement using the variation of focal distance technique by the microscope Alicona™.

5- Calculate the volume of OFHC copper that disappeared from the oxidized surface by treating automatically the images under Matlab® thanks to a script developed for the project making the following tasks:

a) Isolation of green zones (covered by atacamite) starting from a photo of the oxidized surface.

b) Starting from the image of the cleaned surface, isolating topographies of zones corresponding to those which were covered by green before the cleaning.

c) Visualization of the topography corresponding to the filtered zones only. The level 0 corresponds to the mean plan of the freshly machined surface, before the aging test ($R_a < 800$ nm).

d) The program calculates the volume of copper that has disappeared (valleys) under the zones covered with atacamite and divides it by the green analysed surface, which gives a mean value of the substrate in-depth du dissolved as follows:

$$\text{Creux moyen } (\mu\text{m} / mm^2 \text{ vert}) = \frac{\sum_{i \in \text{zones isolées}} \text{profondeur du pixel}_i \times \text{aire du pixel}_i}{\sum_{i \in \text{zones isolées}} \text{aire du pixel}_i} \qquad \textbf{Eq.2}$$

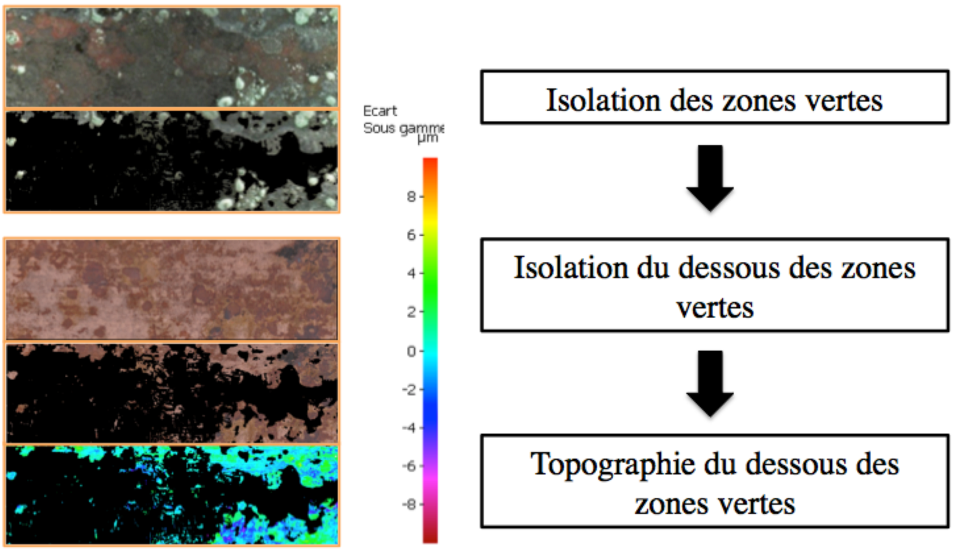


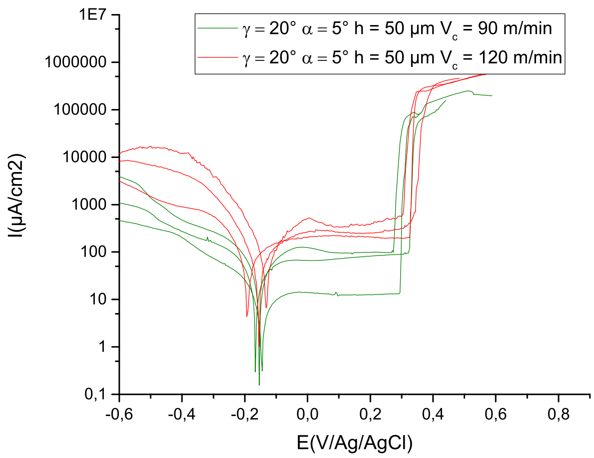
Figure 15. *Evaluation of the volume of copper dissolved by mm² covered by'atacamite.*

I.3 Experimental results

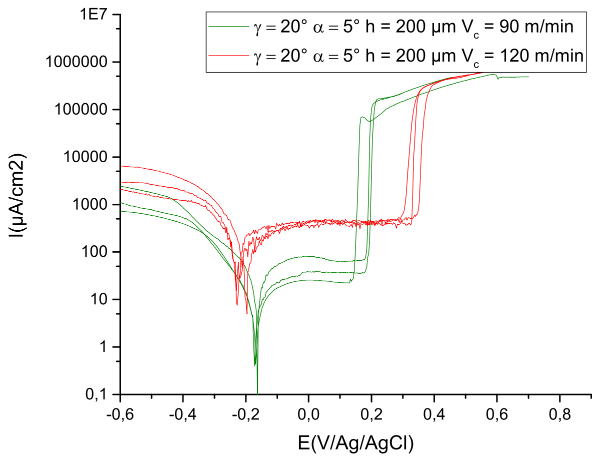
I.3.1 Local electrochemical behavior

a) Influence of the cutting speed V_c

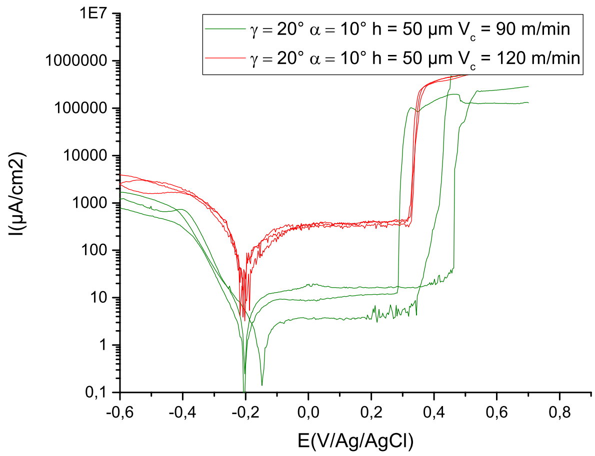
The effects of two cutting speeds 90 et 120 m/min are here compared (cf. Figure 16). In fact, raising the speed resulted in a raise of the passive plate, besides a higher pitting potential when the uncut chip thickness is h = 0.2 mm (cf. Figure 16b).



(a)



(b)



(c)

Figure 16. *Polarization curves for surfaces obtained by orthogonal cutting under the cutting conditions (a) h = 0.05 mm, γ = 20°, α = 5° (b) h = 0.2 mm, γ = 20°, α = 5° (c) h = 0.05 mm, γ = 20°, α = 10°.*

b) Influence of the uncut chip thickness h

To confirm the effect of the uncut chip thickness, polarization curves issued from the analysis of surfaces machined at 90 m/min are compared but they don't show any pertinent effect except for E_b with a rake angle of 20° (there is evident synergy effect between h and γ) (cf. Figure 17a and d).

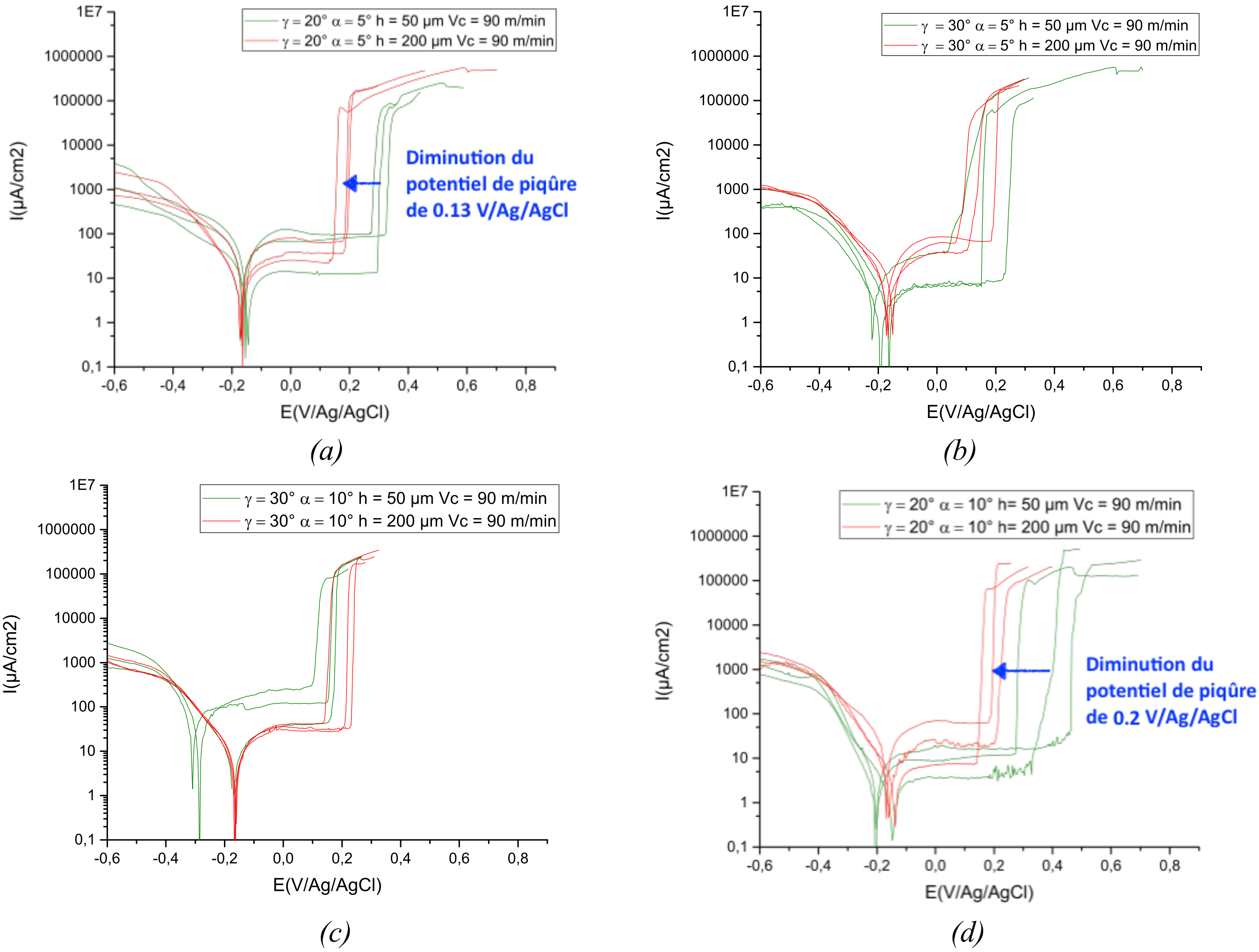


Figure 17. *Polarization curves obtained by analyzing surfaces machined at V_c = 90 m/min.*

I.3.1.1 Summary of the results concerning the electrochemical behavior

All the results related to local electrochemical behavior of the surfaces obtained by orthogonal cutting are gathered in Table 2.

Cutting conditions	**I_{cath} (µA/ cm²)**	**Standard deviation eI_{cath} (µA/cm²)**	**I_{pass} (µA/cm²)**	**Standard deviation eI_{pass} (µA/cm²)**	**E_b (mV/Ag/AgCl)**	**Standard deviation eE_b (mV/Ag/AgCl)**
α = 5° γ = 20° h = 0.05 mm V_c = 90 m/min	2140	1680	55	42	297	26
α = 10° γ = 20° h = 0.05 mm V_c = 90 m/min	1203	455	10	6	372	90
α = 5° γ = 30° h = 0.05 mm V_c = 90 m/min	706	309	21	14	127	89
α = 10°	1718	960	141	102	131	29

$\gamma = 30°$ h = 0.05 mm V_c = 90 m/min						
$\alpha = 5°$ $\gamma = 20°$ h = 0.20 mm V_c = 90 m/min	1559	843	53	28	162	23
$\alpha = 10°$ $\gamma = 20°$ h = 0.20 mm V_c = 90 m/min	1701	607	39	32	185	46
$\alpha = 5°$ $\gamma = 30°$ h = 0.20 mm V_c = 90 m/min	1088	112	61	23	125	59
$\alpha = 10°$ $\gamma = 30°$ h = 0.20 mm V_c = 90 m/min	1230	202	35	7	185	43
$\alpha = 5°$ $\gamma = 20°$ h = 0.05 mm V_c = 120 m/min	7684	4823	294	79	331	9
$\alpha = 10°$ $\gamma = 20°$ h = 0.05 mm V_c = 120 m/min	3263	563	351	34	322	3
$\alpha = 5°$ $\gamma = 20°$ h = 0.20 mm V_c = 120 m/min	4220	2188	438	43	325	24

Table 2. *Results of the local electrochemical analysis of surfaces obtained by orthogonal cutting.*

I.3.2 Corrosion resistance under saline atmosphere

Specimen obtained by orthogonal cutting using a cutting speed of 90 m/min are submitted to aging under saline atmosphere.

Figure 18 shows that the variation of cutting parameters has an influence on the evolution of the corrosion of OFHC copper surfaces. In fact, for surfaces machined with a tool having the same rake angle $\gamma = 20°$, a raise in the uncut chip thickness h accelerates corrosion (cf. Figure 19a and Figure 19b). Nevertheless, this phenomenon is not seen when the rake angle $\gamma = 30°$ where h seems to have no influence. (cf. Figure 19c and Figure 19d). On the other side, whatever the rake angle and the uncut chip thickness are, the raise of the flank angle from 5° to 10° shows a good influence on corrosion resistance.

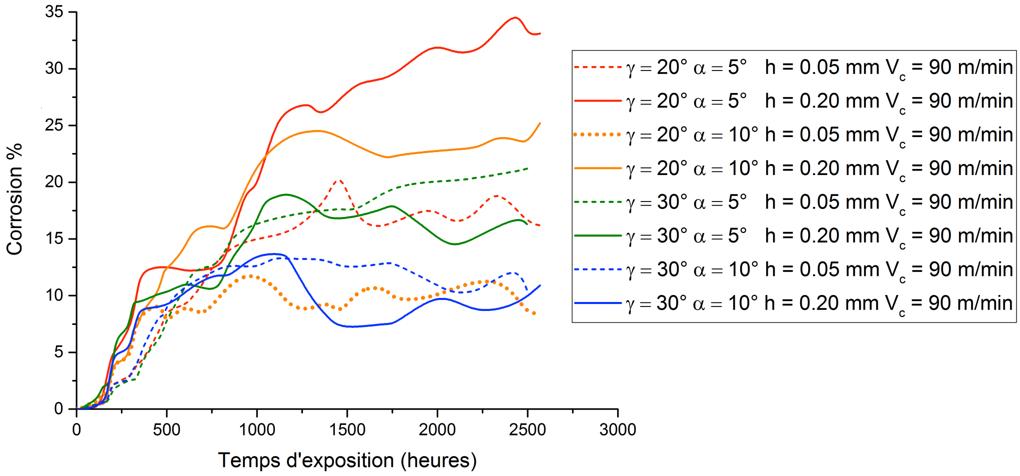


Figure 18. *Corrosion evolution under saline atmosphere concerning surfaces obtained by orthogonal cutting at V_c = 90 m/min.*

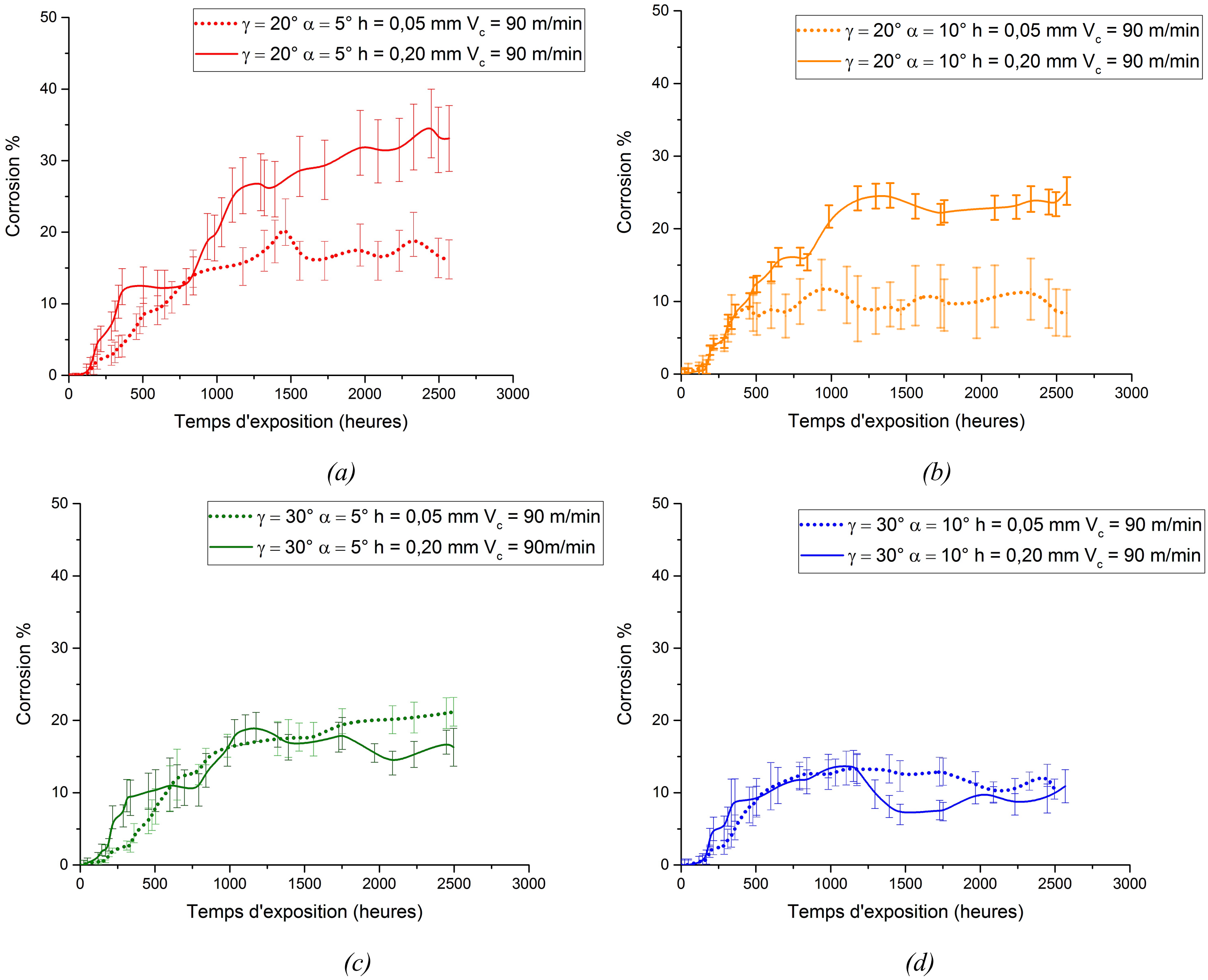


Figure 19. *Influence of the uncut chip thickness on corrosion evolution under saline atmosphere of surfaces obtained by orthogonal cutting.*

The drop of the percentage of surface corrosion as in the case of ($\gamma = 30°$; $\alpha = 10°$; $h = 0.2$ mm et $V_c = 90$ m/min) between 1200 and 1400 aging hours is due to chipping (cf. Figure 20). It consists into the removal of the green surface (atacamite).

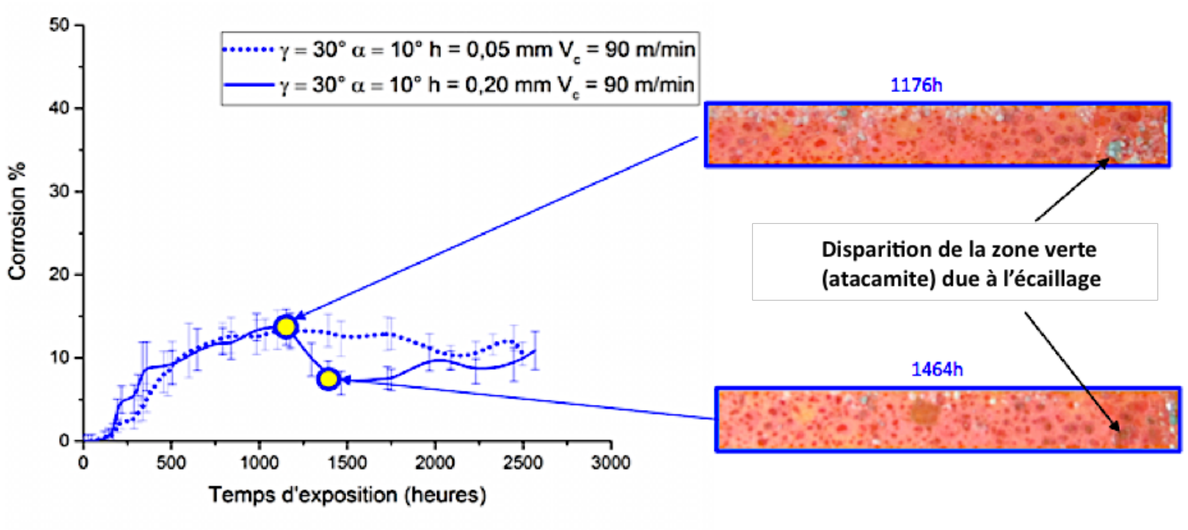


Figure 20. *Green surface removed due to chipping.*

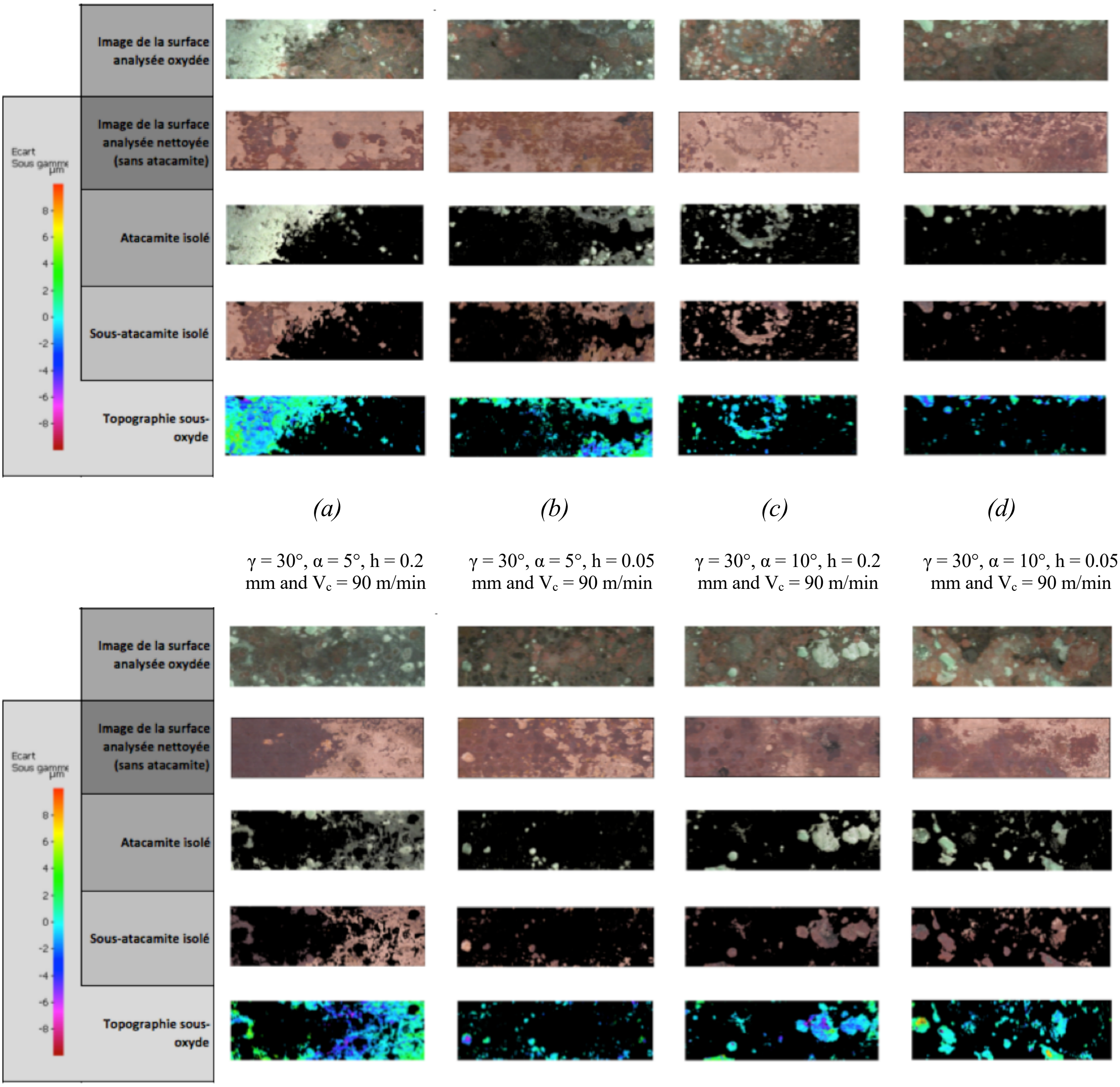
I.3.2.1 Degradation of the machined copper under the green oxide

To explain degradation under the green surface, in-depth topography is measured under the oxide once it is removed, and the measurements results are gathered in Figure 21. The removed volume per surface unit corresponds to the mean depth corroded under the oxide. Results are represented by the Figure 22. Comparing the percentage of oxidized surface at the end of the aging test under saline atmosphere (cf. Figure 23), the following remarks can be cited:

- The less oxidized surface is the less degraded under the green oxide "atacamite" corresponding to the cutting condition ($\gamma = 20°$; $\alpha = 10°$; $h = 0.05$ mm ; $V_c = 90$ m/min),

- The more oxidized surface is not necessarily having the highest in-depth degradation ($\gamma = 30°$; $\alpha = 5°$; $h = 0.05$ mm ; $V_c = 90$ m/min).

A unidirectional profile measured in the oxidized region is taken from every analyzed surface in order to identify the roughness profile under the oxide. It is measured only under the green areas (represented by the green lines in Figure 24). The used in-depth resolution is e 78 nm and the horizontal resolution is 5 µm.



(a) $\gamma = 30°$, $\alpha = 5°$, $h = 0.2$ mm and $V_c = 90$ m/min

(b) $\gamma = 30°$, $\alpha = 5°$, $h = 0.05$ mm and $V_c = 90$ m/min

(c) $\gamma = 30°$, $\alpha = 10°$, $h = 0.2$ mm and $V_c = 90$ m/min

(d) $\gamma = 30°$, $\alpha = 10°$, $h = 0.05$ mm and $V_c = 90$ m/min

(e) $\gamma = 20°$, $\alpha = 5°$, $h = 0.2$ mm and $V_c = 90$ m/min

(f) $\gamma = 20°$, $\alpha = 5°$, $h = 0.05$ mm and $V_c = 90$ m/min

(g) $\gamma = 20°$, $\alpha = 10°$, $h = 0.2$ mm and $V_c = 90$ m/min

(h) $\gamma = 20°$, $\alpha = 10°$, $h = 0.05$ mm and $V_c = 90$ m/min

Figure 21. *In-depth topography of the specimen surfaces affected by the corrosion.*

According to the obtained profiles, it is remarkable that under the green areas two types of cavities can prevail: large cavities with plate bottom (generalized dissolution) and thin and deep cavities (intergranular corrosion) in the borders of the green area (cf. Figure 24).

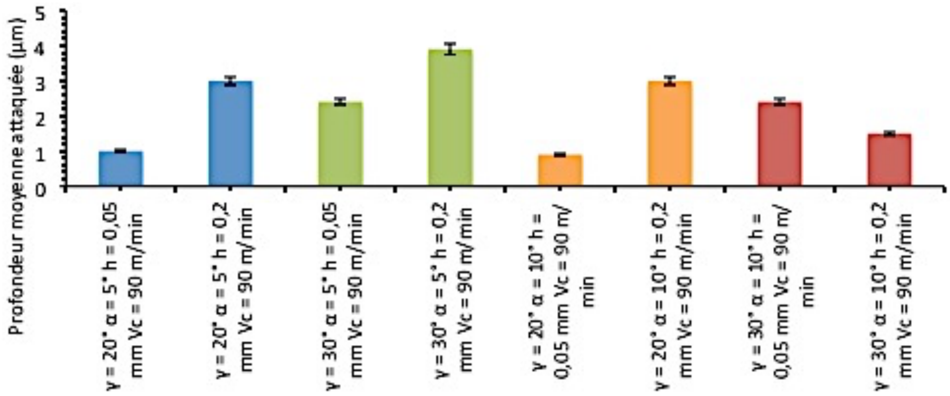


Figure 22. *Mean depth degraded under the green areas.*

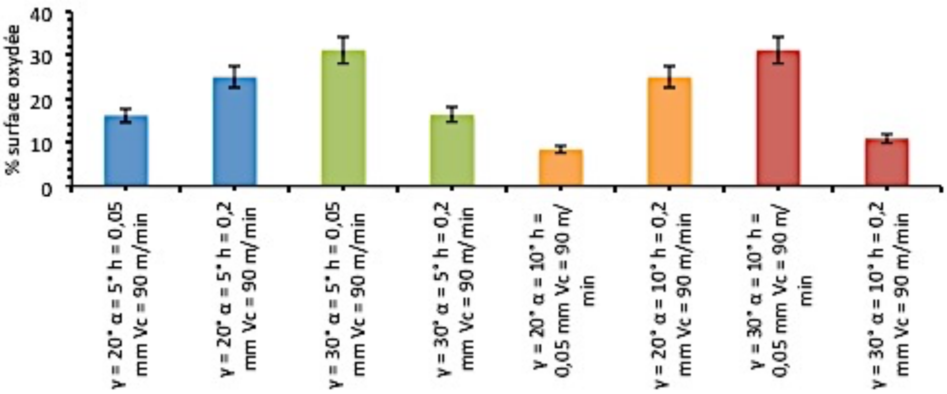
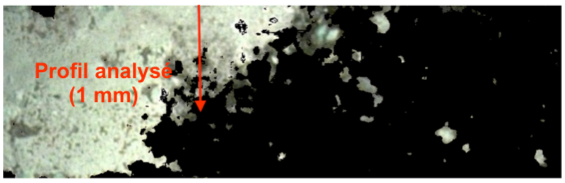
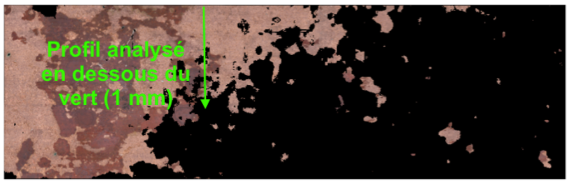


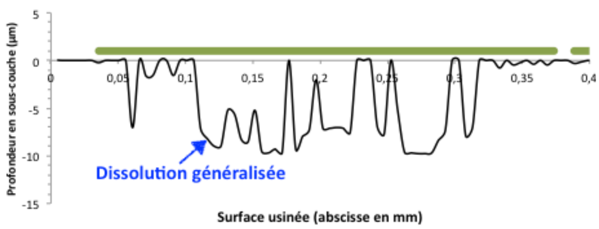
Figure 23. *Percentage of the oxidized surfaces at the end of the aging test under saline atmosphere.*



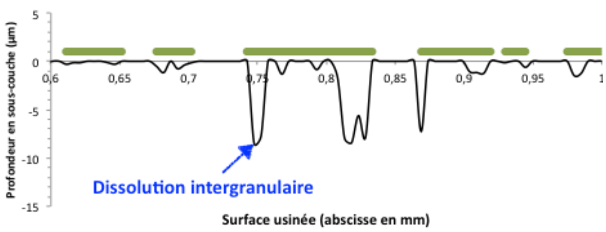
(a) Surface covered by green oxide



(b) Surface under the green oxide



(c) Part of the analyzed profile presenting a generalized dissolution.



(d) Part of the analyzed profile presenting intergranular corrosion.

Figure 24. *Profils sous les zones vertes des surfaces usinées en rabotage pour $\gamma = 30°$, $\alpha = 5°$, $h = 0.2$ mm et $V_c = 90$ m/min.*

The depth of the cavities vary under the surfaces covered by the green oxide, what witnesses a non uniform dissolution happening on the same surface. These observations under the oxides allow the corrosion process analysis explaining the surface damage in the presence of saline atmosphere.

Inside the droplets, copper dissolution leads to the formation of the ions Cu^{+} or Cu^{2+} in aqueous solution. Associated to anions Cl^{-} already present in the droplet, they constitute the green compound CuCl. With high chloride concentration, they can even produce the ions of $CuCl^{2-}$. On the other side, cations Cu^{2+} can be hydrolyzed to form atacamite $Cu_2(OH)_3Cl$. This reaction, consuming molecules of H_2O, produces protons H^{+}, which explains the drop of pH. As the droplet stagnates in the surface, the pH diminishes with the generation of atacamite. A previous study (Gravier 2009) has demonstrated that pH lower than the limit of copper dissolution can be rapidly reached leading to a generalized dissolution and to intergranular corrosion. The electrons issued from the dissolution reduce the oxygen inside the droplet. As the solubility of oxygen inside water is high, the cathodic reaction continues (cf. Figure 25).

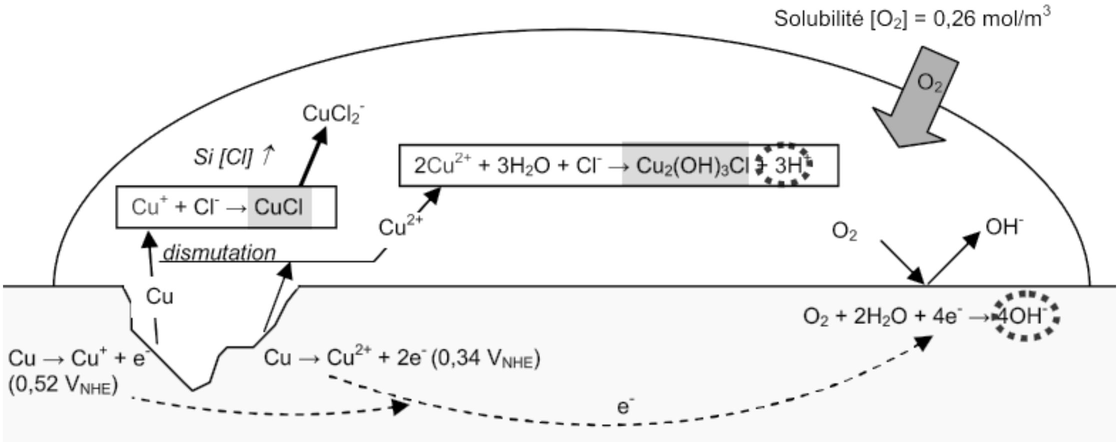


Figure 25. *Degradation model of copper surfaces (Gravier 2009).*

I.4 Relationship between cutting conditions – surface integrity – corrosion

In this paragraph, all experimental and numerical results concerning surface integrity, local electrochemical behavior and corrosion under saline atmosphere are treated under a statistical approach in order to identify:

1 – What are the surface integrity parameters that have relevant impact on corrosion evolution besides electrochemical behavior of the surfaces? What is their importance ranking? What are the activated mechanisms?

2 – For those surface integrity parameters identified as relevant, what are the cutting conditions that master them?

I.4.1 Statistical approach

For results analysis, two kinds of statistical treatments are performed: correlation analysis in addition to the analysis of variance.

The statistical approach is explained in the following figure:

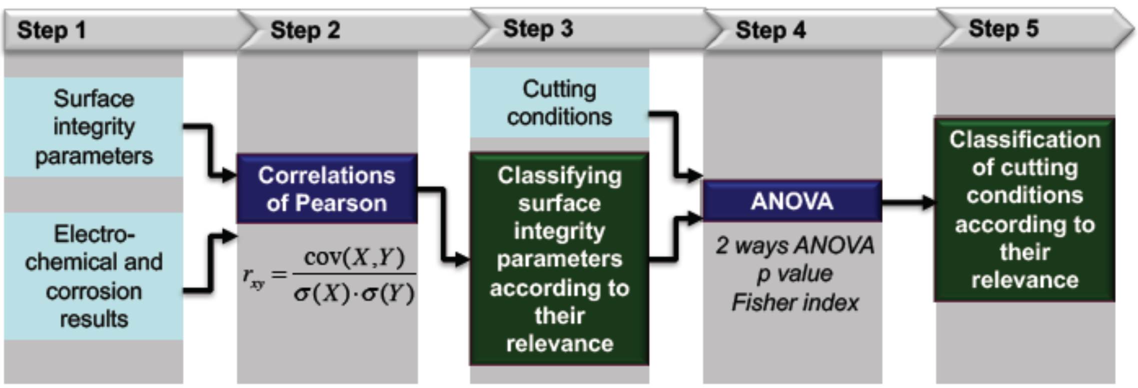


Figure 26. *Multi-physical analysis approach.*

I.4.1.1 Pearson's correlation analysis

Correlation analysis, called the correlation of Bravais-Pearson, consists in studying the correlation between two or more statistical variables or random entities: it is a quantification of the relationship potentially existing between two variables by the coefficient r_{ay} calculated as follows:

$$r_{xy} = \frac{\mathrm{cov}(X,Y)}{\sigma(X)\cdot\sigma(Y)} \qquad \textbf{Eq. 3}$$

with σ the standard deviation and $cov(X,Y)$ the covariance of the variables X and Y. For this study, the variable X corresponds to a parameter of the electrochemical behavior -corrosion (current density in the cathodic domain I_{cath}, its corresponding dispersion eI_{cath} (standard deviation related to 3 trials performed on each tested surface), current density in the passive plate I_{pass}, its corresponding dispersion eI_{pass}, pitting potential E_b, its corresponding dispersion eE_b, percentage of the oxidized surface and the mean oxidized depth under the oxide by the end of the aging test in saline atmosphere). The variable Y corresponds to a parameter of the surface integrity (residual stress in the cutting direction σ_x, transversal residual stress σ_y, plastic layer thickness, surface arithmetic roughness S_a, surface roughness at maximum height S_t, grain size at the surface d_{grain}, dislocation density at the surface ρ, and micro-hardness $HV_{0.025}$). The correlation coefficient is unitless, what allows the comparison between the variables. Each measurement is normalized, and defined between -1 and 1. The correlation coefficient aims firstly to characterize a linear relationship, either positive or negative. The more it is near 1 (absolute value), the more the relationship is strong. A zero r_{xy} signifies the absence of any correlation.

I.4.1.2 Analysis of variance (ANOVA)

The analysis of variance aims to study the behavior of a continuous variable in function of one or many categorical variables. In this study, the continuous variable corresponds to one of the surface integrity parameters identified as relevant and the categorical variable corresponds to a cutting condition (cutting speed V_c, uncut chip thickness h, flank angle α, and rake angle γ). The object is to know which continuous variable has values significantly different according to each category (i.e., according to the factor level, here the cutting condition, of the interaction between two factors). More specifically, the null hypothesis is tested: there are at least two levels for which the mean value of the variable of surface integrity shows difference. In the performed study, the significance of the analysis is verified by a generalized Fisher test. According to the Fisher index, variables can be classified according to their level of importance.

I.4.1.3 Steps of the analysis

The first step consists in gathering all the experimental and numerical measured values concerning corrosion (Table 3) and surface integrity (Table 4) in function of the order of the cutting conditions given by Table 5.

The experimental results corresponding to residual stresses, grains size, dislocation density and micro-hardness cover only a part of the design of experiments. To get the full DOE, results issued from numerical simulations are used.

Test	I_{cath}	I_{pass}	E_b	eI_{cath}	eI_{pass}	eE_b	Oxidized surface	Mean oxidized depth
N°	µA/cm²	µA/cm²	mV/Ag/AgCl	µA/cm²	µA/cm²	mV/Ag/AgCl	%	µm
1	7684	294	331	4823	79	9	-	-
2	3263	351	322	563	34	3	-	-
3	4220	438	325	2188	43	24	-	-
4	1559	53	162	843	28	23	33.1	3
5	1718	141	131	960	102	29	10.5	2.4
6	2140	55	297	1680	42	26	16.2	1
7	1230	35	185	202	7	43	10.9	1.5
8	1702	39	185	607	32	46	25	3
9	1088	61	125	112	23	59	16.3	3.9
10	706	21	127	309	14	89	31.2	2.4
11	1203	10	372	455	6	90	8.4	0.9

Table 3. *Results issued from local electrochemical analysis and aging under saline atmosphere.*

Test	σ_x	σ_y	Plastic layer thickness	S_a	S_t	d_{grain}	ρ	Micro-hardness
N°	MPa	MPa	µm	nm	µm	µm	µm/µm³	$HV_{0.025}$
1	228	145	61	471	22.6	9.7	1350	122
2	188*	148*	214	493	5.6	10	1270	122
3	137	116	777	495	18.1	8.2	1900	128
4	106	81	678	626	17.4	7.7	2130	130
5	134*	118*	195	672	31	14	1370	114
6	181*	145*	44	471	22.6	5.4	4400	146
7	59*	50*	362	662	15.3	6.1	3370	139
8	-10*	9*	794	677	16.9	7.1	2590	134
9	36*	41*	384	461	20.9	5.2	4770	148
10	204	105	151	440	18.8	3.1	1350	186
11	206	130	149	550	5.7	8	1900	128

**experimental data*

Table 4. *Results issued from experimental and numerical analysis of the surface integrity.*

Test	γ	α	h	V_c
N°	°	°	mm	m/min
1	20	5	0.05	120
2	20	10	0.05	120
3	20	5	0.2	120
4	20	5	0.2	90
5	30	10	0.05	90

6	20	5	0.05	90
7	30	10	0.2	90
8	20	10	0.2	90
9	30	5	0.2	90
10	30	5	0.05	90
11	20	10	0.05	90

Table 5. *Cutting conditions while orthogonal cutting.*

The second step consists into calculating Pearson's correlations for each pair (surface integrity parameter / corrosion parameter). Results are presented in Table 6. It is worth to remind that a positive value of the correlation coefficient corresponds to an evolution of the pair in the same direction, and that a negative value corresponds to opposite evolution.

	σ_x	σ_y	S_a	S_t	d_{grain}	ρ	$HV_{0.025}$	Plastic layer
I_{cath}	0.43	0.47	-0.32	0.12	0.38	-0.24	-0.49	0.33
I_{pass}	0.32	0.48	-0.32	-0.05	0.43	-0.37	-0.42	0.27
E_b	0.57	0.64	-0.34	-0.51	0.13	-0.14	-0.37	-0.12
eI_{cath}	0.46	0.48	-0.30	0.32	0.23	-0.05	-0.28	0.24
eI_{pass}	0.21	0.37	0.11	0.70	0.77	-0.23	-0.49	-0.04
eE_b	-0.05	-0.29	-0.08	-0.20	-0.50	0.02	0.60	-0.26
Oxidized surface	-0.07	-0.28	-0.35	0.09	-0.42	-0.24	0.49	0.50
Mean oxidized depth	-0.63	-0.66	0.03	0.35	-0.02	0.10	0.09	0.62

Table 6. *Pearson's correlations between parameters of corrosion-electrochemical behavior and parameters of surface integrity.*

The third step consists in identifying surface integrity parameters having correlation coefficients higher than 0.5 (absolute value).

In the fourth step, selected parameters become the object of the analysis of variance in function of cutting conditions and their double interactions for synergy effects revelation. So, Fisher indexes are calculated and dissociation probabilities *p* are revealed (Table 7). A probability *p* lower than 0.2 is accepted.

In the fifth step, according to Fisher indexes values (Table 8), a ranking according to the relevance is performed. All calculations are done using the software Minitab Express^{TM}.

	γ	α	h	V_c	V_c×h	V_c×α	h×α	h×γ	α×γ
E_b	0.0079	0.8590	0.2710	0.0410	0.5880	0.7170	0.7540	0.0914	0.8550
eE_b	0.2160	0.8390	0.9160	0.0420	0.3790	0.6830	0.8230	0.8650	0.0850
I_{cath}	0.1360	0.4090	0.5300	0.0013	0.5210	0.0970	0.8260	0.7030	0.3400
eI_{cath}	0.1740	0.1980	0.4450	0.0330	0.9570	0.0580	0.5800	0.9180	0.1680
I_{pass}	0.2560	0.6980	0.8390	0.0001	0.0750	0.7700	0.3870	0.8830	0.5650

eI_{pass}	0.9510	0.9190	0.2980	0.3370	0.9140	0.4400	0.7340	0.3770	0.1600
Oxidized surface	0.9250	0.2740	0.9100	-	-	-	0.3190	0.2670	0.4160
Mean oxidized depth	0.4853	0.4462	0.1209	-	-	-	0.4417	0.2188	0.5343
S_a	0.7732	0.0303	0.2508	0.2032	0.6643	0.2207	0.7205	0.4601	0.1155
S_t	0.2098	0.2635	0.9994	0.5536	0.6347	0.2688	0.6570	0.3091	0.0940
d_{grain}	0.6400	0.1680	0.4220	0.2810	0.9050	0.5440	0.1559	0.5900	0.1500
HV	0.1820	0.1900	0.9660	0.2240	0.7049	0.4200	0.2150	0.7780	0.0780
Plastic layer thickness	0.5415	0.9724	0.0006	0.9766	0.3421	0.5845	0.5183	0.0010	0.9714
σ_x	0.4452	0.5102	0.0014	0.2004	0.3049	0.7421	0.4478	0.9798	0.9201
σ_y	0.3042	0.6389	0.0032	0.1114	0.1678	0.6325	0.1758	0.8512	0.5850

Table 7. *Probability of dissociation p.*

	γ	**α**	**h**	**V_c**	**$V_c \times h$**	**$V_c \times \alpha$**	**$h \times \alpha$**	**$h \times \gamma$**	**$\alpha \times \gamma$**
E_b	11.6	0.0	1.4	5.7	0.3	0.1	0.1	3.8	0.0
eE_b	1.8	0.0	0.0	5.6	0.9	0.2	0.1	0.0	4.0
I_{cath}	2.7	0.8	0.4	21.2	0.5	3.7	0.1	0.2	1.0
eI_{cath}	2.2	1.9	0.6	6.3	0.0	5.1	0.3	0.0	2.4
I_{pass}	1.5	0.2	0.0	86.4	4.4	0.1	0.9	0.0	0.4
eI_{pass}	0.0	0.0	1.2	1.0	0.0	0.7	0.1	0.9	2.5
Oxidized surface	0.0	1.5	0.0	-	-	-	1.4	1.8	0.9
Mean oxidized depth	0.6	0.7	3.3	-	-	-	0.7	2.1	0.5
S_a	0.1	6.6	1.5	1.9	0.2	1.8	0.1	0.6	3.2
S_t	1.8	1.4	0.0	0.4	0.3	1.4	0.2	1.2	3.8
d_{grain}	0.2	2.2	0.7	1.3	0.0	0.4	2.5	0.3	2.6
HV	2.1	2.0	0.0	1.7	0.2	0.7	1.9	0.1	4.3
Plastic layer thickness	0.4	0.0	26.0	0.0	1.0	0.3	0.5	29.6	0.0
σ_x	0.6	0.5	20.6	1.9	1.2	0.1	0.7	0.0	0.0
σ_y	1.2	0.2	15.8	3.1	2.4	0.3	2.3	0.0	0.3

Table 8. *Indices de Fisher.*

I.4.2 Results of the statistical analysis

Correlations analysis permitted the identification of the influent parameters regarding electrochemical behavior and corrosion. Figure 27 shows that, for surfaces corrosion during long term aging tests, first the thickness of the plastic layer plays the most important role towards the oxidized surface area, and second it is the micro-hardness.

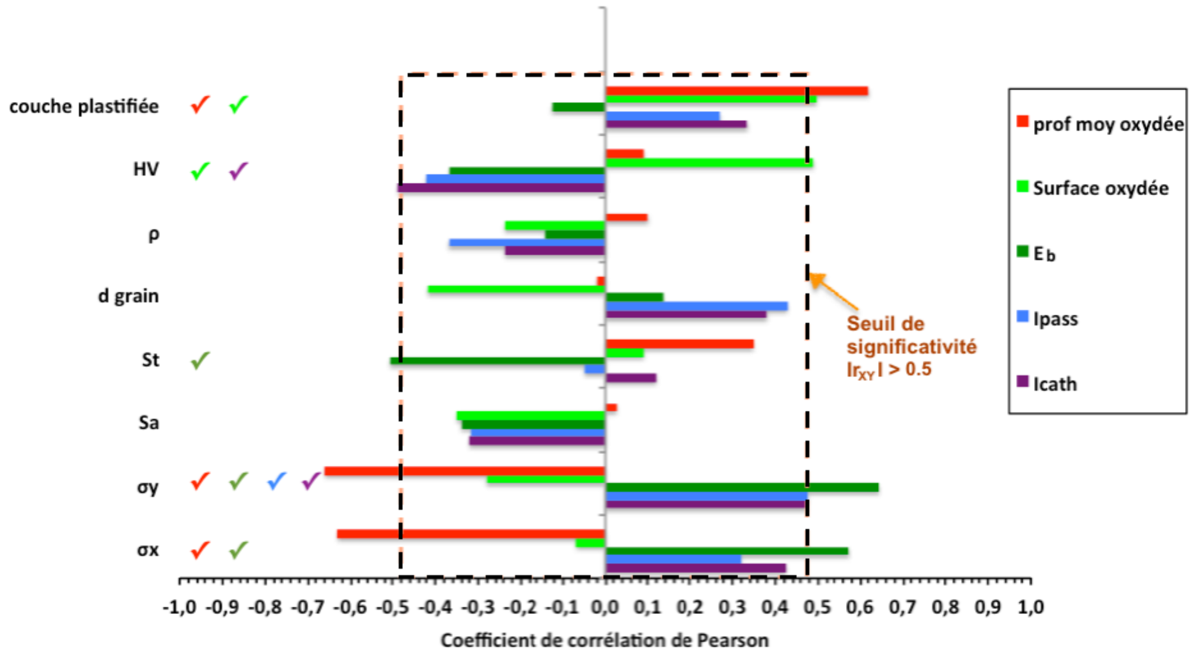


Figure 27. *Pearson's correlation coefficients concerning the parameters of corrosion and local electrochemical behavior.*

However, in what concerns the in-depth oxidation, residual stresses seem to be the most relevant parameters, followed by the plastic layer thickness.

Concerning to local electrochemical behavior and the parameters issued from polarization curves, it is clear that only the residual stress σ_y and the micro-hardness *HV* are significantly influents on current density in the cathodic domain I_{cath} (cf. Figure 27) besides the dispersion of the measured values (reflecting the repeatability of the process) eI_{cath} (cf. Figure 28). Concerning the passivation of the surface, only the residual stress σ_y seems significantly influent on I_{pass} (cf. Figure 27), however, the repeatability of the measurement eI_{pass} depends firstly on the grains size d_{grain} then the roughness at maximal profile height S_t (cf. Figure 28). Finally, for local pitting corrosion, it has been shown that pitting potential E_b depends in a first order on the residual stresses σ_x then σ_y, and in a second order on the roughness S_t (cf. Figure 27). Nevertheless, its dispersion depends on the micro-hardness and the grains size d_{grain} (cf. Figure 28).

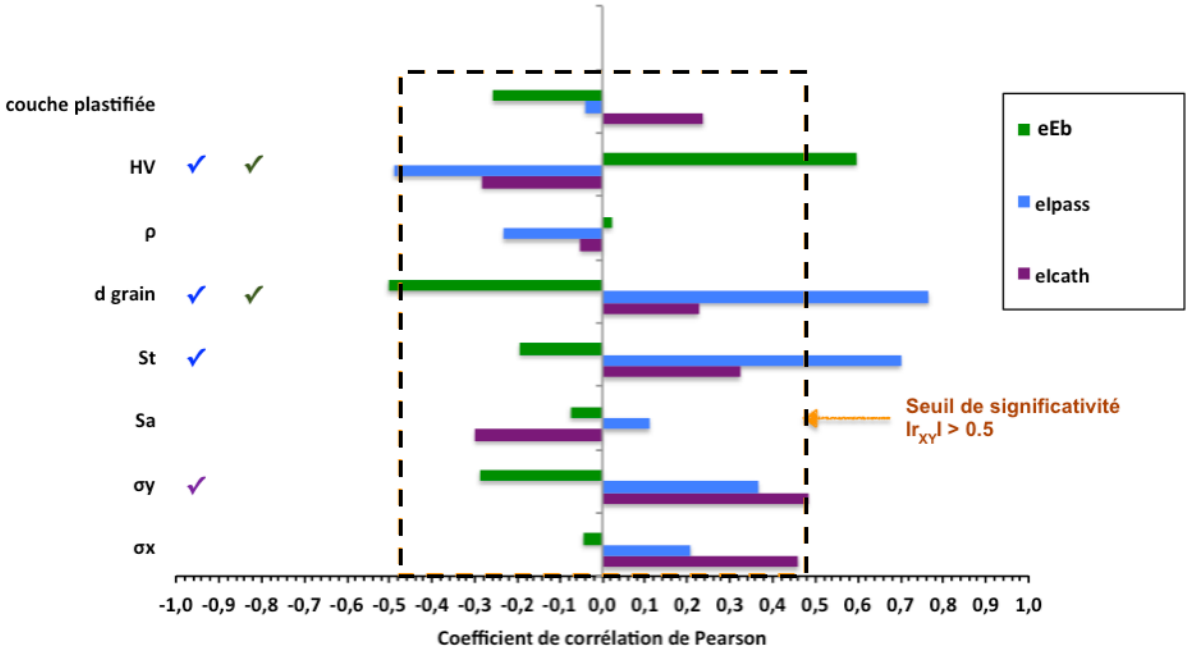


Figure 28. *Pearson's correlation coefficients concerning the dispersions of the parameters related to the electrochemical behavior.*

Definitely, the parameters kept as factors of real impact on corrosion resistance and electrochemical reactivity of the OFHC copper surfaces are gathered in Table 9 (classified by their influence ranking).

Ranking	Parameters of surface integrity	Electrochemical behavior affected by the parameter	Dispersions affected by the parameter
1st	Contrainte résiduelle σ_y	Profondeur moyenne oxydée, E_b, I_{pass} et I_{cath}	eI_{cath}
2nd	Microdureté HV	Surface oxydée et I_{cath}	eI_{pass} et eE_b
3rd	Contrainte résiduelle σ_x	Profondeur moyenne oxydée, E_b	
3rd	Epaisseur de la couche plastifiée	Profondeur moyenne oxydée et la surface oxydée	
4th	Rugosité à hauteur maximale de profil S_t	E_b	eI_{pass}
5th	taille des grains d_{grain}		eI_{pass} et eE_b

Table 9. *Ranking of the parameters infuencing the electrochemical behavior - corrosion of the machined surfaces.*

After the analysis of variance « 2 ways ANOVA » between the identified parameters and the cutting conditions besides their double interactions, the probabilities p corresponding to the hypothesis of dissociation of the factors showed the results given in Figure 29, in particular:

- The longitudinal residual stress σ_x is associated to the uncut chip thickness h, and the transversal residual stress σ_y is associated to the uncut chip thickness h, the cutting speed V_c, their interaction ($V_c \times h$), and to the interaction between the flank angle / the uncut chip thickness ($\alpha \times h$).

- The thickness of the plastic layer is associated to the uncut chip thickness h and to the interaction rake angle / uncut chip thickness ($\gamma \times h$).

- The roughness S_t is associated to the interaction rake angle / flank angle ($\gamma \times \alpha$).

- The micro-hardness and the grains size are associated to the interaction ($\gamma \times \alpha$), then ($h \times \alpha$), and finally to the angles α and γ.

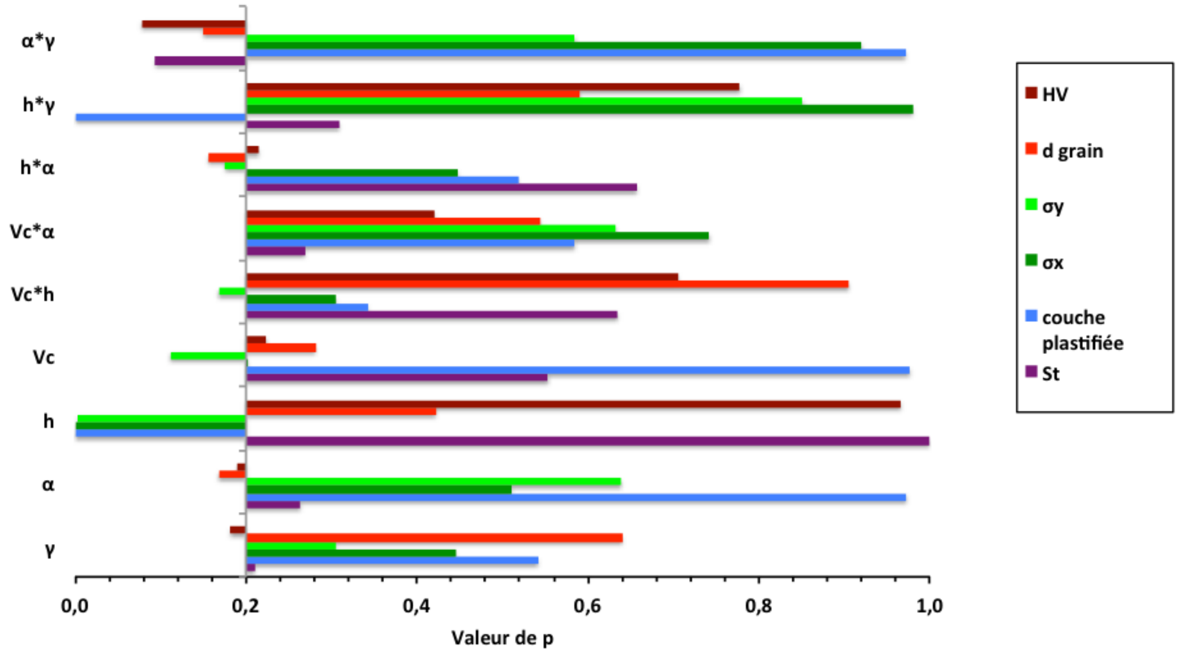


Figure 29. *Probabilité p de dissociation calculée (2 ways ANOVA) pour les différents résultats de l'intégrité de surface.*

To quantify the relevance of these parameters and their interactions, Fisher indexes F_i (cf. Figure 30) allow the ranking basing on the following criteria:

- $F_i \geq 25$: 1st order parameter.
- $2.5 < F_i \leq 25$: 2nd order parameter.
- $F_i \leq 2.5$: 3rd order parameter.

The results of this ranking are as follows:

1- The uncut chip thickness h (order 1).

2- The interaction uncut chip thickness with the rake angle ($h \times \gamma$) (order 1).

3- The interaction flank angle with rake angle ($\alpha \times \gamma$) (order 2).

4- The cutting speed V_c (order 2).

5- The interaction uncut chip thickness with the flank angle ($h \times \alpha$) (order 2).

6- The interaction cutting speed with the uncut chip thickness ($V_c \times h$) (order 3).

7- The flank angle α (order 3).

8- The rake angle γ (order 3).

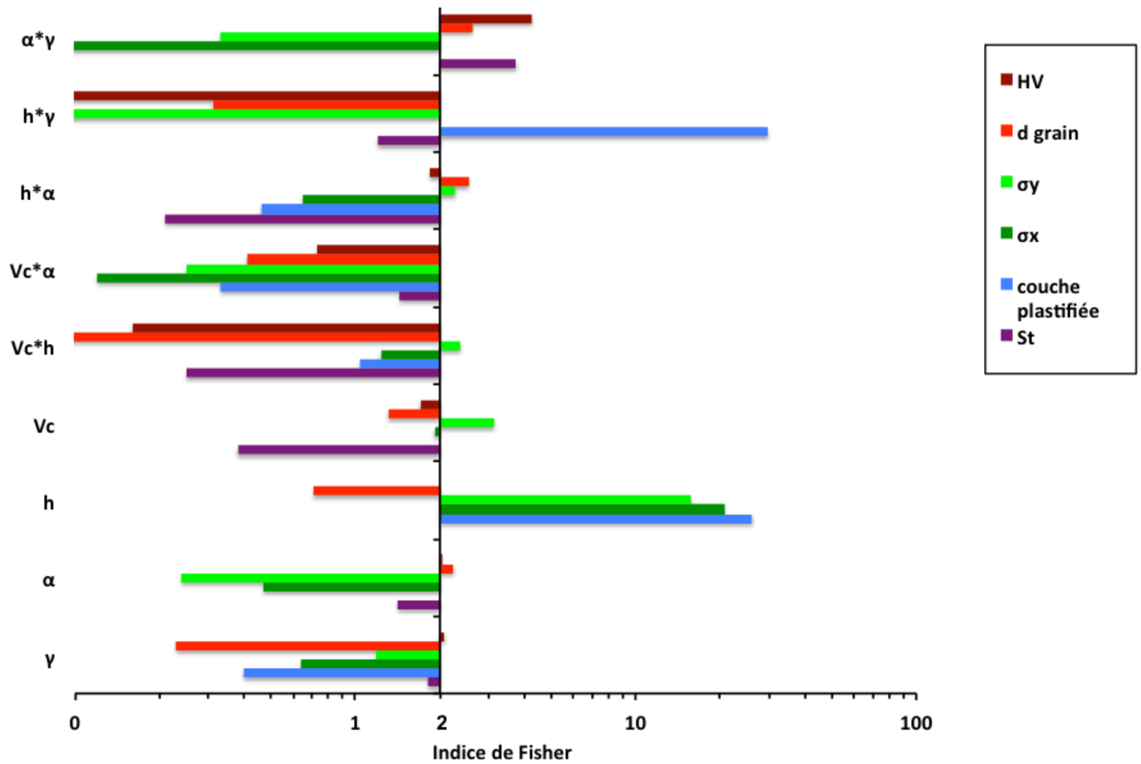


Figure 30. *Fisher indexes resulting from the analysis of variance to establish the relations betweenthe surface integrity parameters and the cutting conditions with consideration of the eventual interactions effects during orthogonal cutting.*

Although relations between the different parameters are identified, the physical origins of those relations should be clarified, and it is the topic of the next paragraph.

I.5 Physical analysis of the impact of cutting conditions on surface integrity and their consequences on corrosion resistance

The statistical analysis has highlighted the impact of machining on surface integrity, which has shown consequences on corrosion resistance. In this paragraph, a physical explanation of the discovered relations is presented.

I.5.1 Physical origins of the impact of cutting conditions on surface integrity

In order to understand the physical origins of the existing correlations between the cutting conditions and surface integrity parameters identified as relevant for corrosion resistance, the relations are discussed one by one as follows:

I.5.1.1 Relation between the residual stresses and the cutting conditions

The longitudinal residual stress measured on the surface in the cutting direction σ_x is associated to the uncut chip thickness h (it diminishes with the raise of h as shown in Figure 31). However, the residual stress in the transversal direction parallel to the cutting edge σ_y is associated not only to the uncut chip thickness h (diminish also with the raise of h) but also to the cutting speed V_c (raises with V_c), besides their interaction $V_c \times h$ (synergy effect).

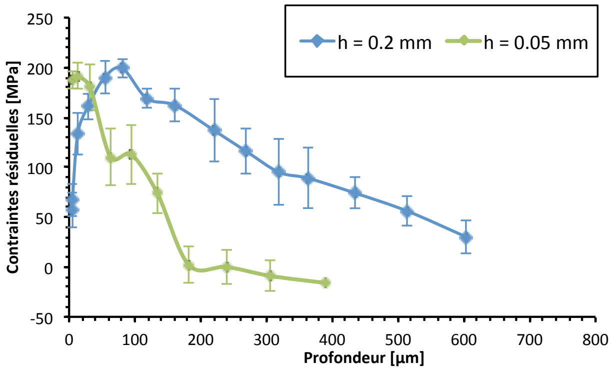
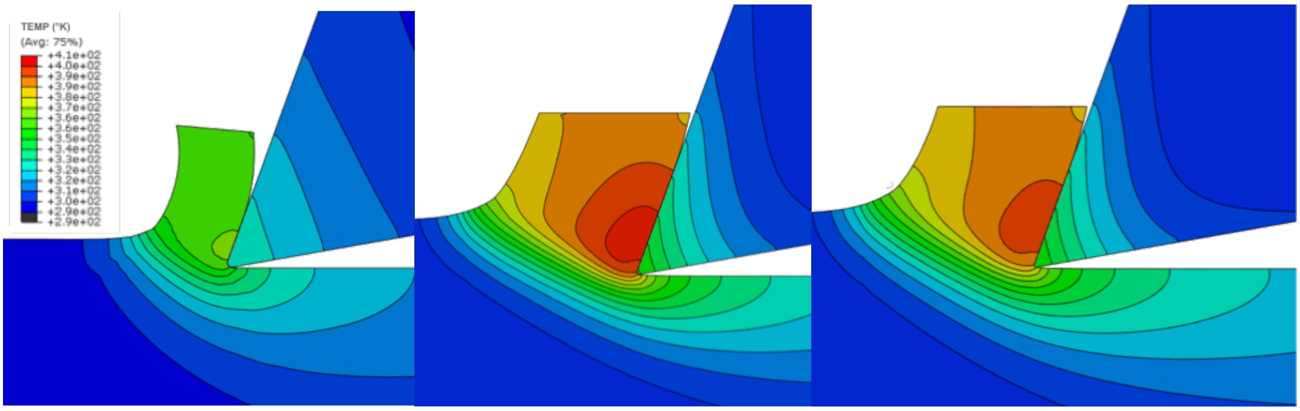


Figure 31. *Longitudinal residual stress σ_x with the variation of the uncut chip thickness h (γ = 20°; α = 10° ; V_c = 120 m/min).*

In fact, residual stresses are the result of mechanical and thermal charges taking place around the cutting edge. Starting from that point, two hypotheses are to be verified to explain the obtained observations: the influence of temperatures distribution and the influence of the state of stress, in particular triaxiality.

The impact of h and V_c on the distribution of the cutting temperatures can be checked by numerical simulation (cf. Figure 32). In fact, this figure shows that, for an uncut chip thickness h of 0.05 mm, the maximal temperature at the generated surface is too low, of 360°K (87°C).

When h moves to 0.2 mm at the same speed, it raises up reaching 390°K (117°C) (cf. Figure 32a and Figure 32b). The same way, a raise in the cutting speed V_c from 90 to 120 m/min leads to a raise in temperatures by moving from 380°K (107°C) to 390°K (117°C) (cf. Figure 32c and Figure 32b), but less important. In fact, between both temperatures, the yield stress varies only of about 1.7%.



(a) *(b)* *(c)*

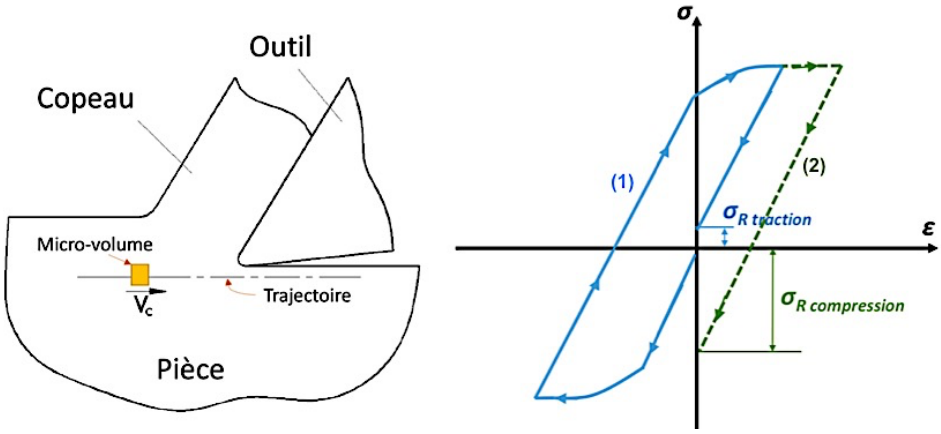
Figure 32. *Chart of the simulated temperatures while orthogonal cutting in °K for : (a) $\gamma = 20°$, $\alpha = 10°$, $h = 0.05$ mm and $V_c = 120$ m/min ; (b) $\gamma = 20°$, $\alpha = 10°$, $h = 0.2$ mm and $V_c = 120$ m/min ; (c) $\gamma = 20°$, $\alpha = 10°$, $h = 0.2$ mm and $V_c = 90$ m/min*

As the temperature variation from a cutting condition to another one is still not significant enough, residual stresses are then essentially generated by mechanical loadings induced by the tool.

To understand the mechanism of residual stresses formation due to machining, the approach initially proposed by Liu and Barash (1976) then by Matsumoto (1999) is used. According to Liu and Barash (1976), a simple method to describe the deformation process induced mu machining consists in using the stress – strain curves obtained by a standard tensile uniaxial test. So, in-depth residual stresses are the consequence of a compressive loading followed by a tensile loading applied to a micro-volume of the workmaterial moving with the speed V_c, when it follows the path represented by Figure 116. This figure shows the level reached by the residual stresses after a loading cycle. According to the predominant loading type, the residual stress will be tensile or compressive. This representation supposes that the strain in the considered path direction (the cutting direction) does not change after a loading cycle.

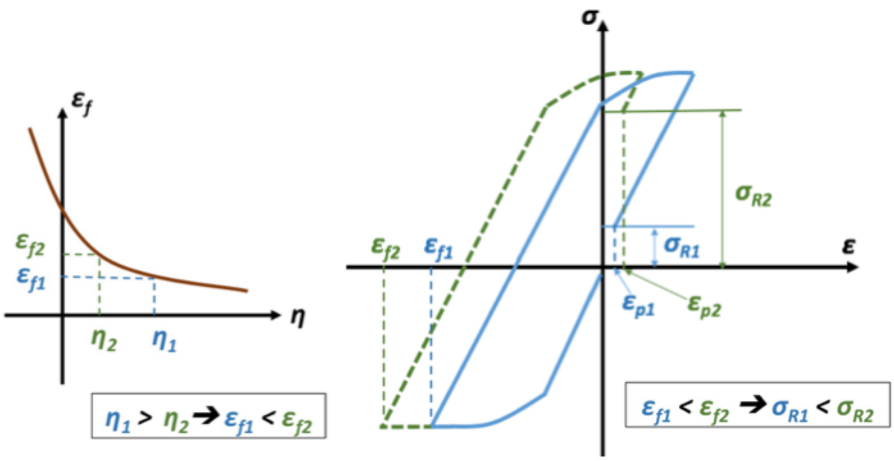
As the work of the plastic strain up to fracture depends on the state of triaxiality of the stresses in the deformation zone, it can be suggested that this variation of the stresses triaxiality (and so the energy required for cutting) will influence surface integrity, including the residual stresses.

In fact, the influence of stresses triaxiality on the equivalent strain at damage initiation (cf. Figure 117a) and then on the residual stresses (cf. Figure 117b) may be explained by a representation of two loading cycles resulting from two different triaxialities. A correction is given to the representation of Liu and Barash in this scheme. It consists in taking into consideration the residual plastic strains ε_p. The highest plastic strain is obtained for the loading cycle presenting the highest compression (Outeiro 2002).



(a) (b)

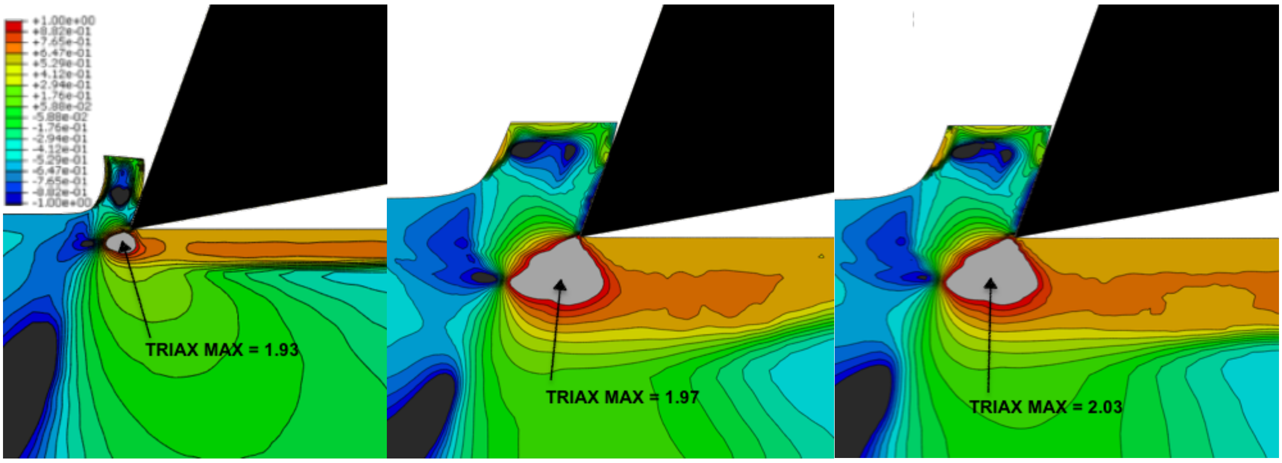
Figure 33. *(a) Micro-volume of the considered workmaterial and its path during cutting. (b) Formation of residual stresses at the level of the considered micro-volume: 1) predominant compressive loading, 2) predominant tensile loading.*



(a) (b)

Figure 34. *(a) Influence of the stresses triaxiality η on the equivalent strain at damage initiation ε_f. (b) Stress-strain diagram showing the influence of ε_f on the formation of residual stresses σ_R dues to the tool's mechanical action on the surface.*

According to the proposed diagram, the falling triaxiality η implies a raise of ε_f, consequently the residual stress σ_R is found more tensile. If it is a tensile residual stress, it would become higher, as shown in Figure 34b. In fact, simulations demonstrated a raise in the triaxiality while the drop of the cutting speed V_c in the deformation zone (cf. Figure 35b and c). It shows also a clear raise in triaxialities while raising the uncut chip thickness *h* (cf. Figure 35a et b), more precisely at the separation point of the chip from the workpiece, what implies a decrease of the strain at the damage initiation of the workmaterial. This induced a decrease of the cutting energy, and consequently the residual stresses are decreased.



(a) *(b)* *(c)*

Figure 35. *Charts of simulated stresses triaxialities during orthogonal cutting for: (a) $\gamma = 20°$, $\alpha = 10°$, $h = 0.05$ mm and $V_c = 120$ m/min ; (b) $\gamma = 20°$, $\alpha = 10°$, $h = 0.2$ mm and $V_c = 120$ m/min ; (c) $\gamma = 20°$, $\alpha = 10°$, $h = 0.2$ mm and $V_c = 90$ m/min.*

I.5.1.2 Relation between the plastic layer under the machined surface and the cutting conditions

The analysis of variance has shown that the thickness of the plastic layer under the machined surface is associated to the uncut chip thickness *h* besides its interaction with the rake angle ($\gamma \times h$). This effect is seen in the in-depth plastic strain profiles.

To model what happens in-depth, 2 micro-volumes of the workmaterial are considered and their loading cycles due to the action of the tool on the surface are drawn as represented in Figure 36. The difference between the two micro-volumes is the value of stresses triaxiality induced by the cutting geometry. The loading cycle represented in bleue in Figure 34b corresponds to the micro-volume 1, and the green one to micro-volume 2. The increase of the rake angle γ leads to a clear increase in stresses triaxiality η (cf. Figure 37), so to a decrease of the equivalent strain at damage initiation ε_f, then to a drop in the plastic strain value ε_p.

The effect of the rake angle γ on stresses triaxiality and consequently on the plastic layer is confirmed by the investigations of Abushawashi (2013) about AISI 1045 machining process.

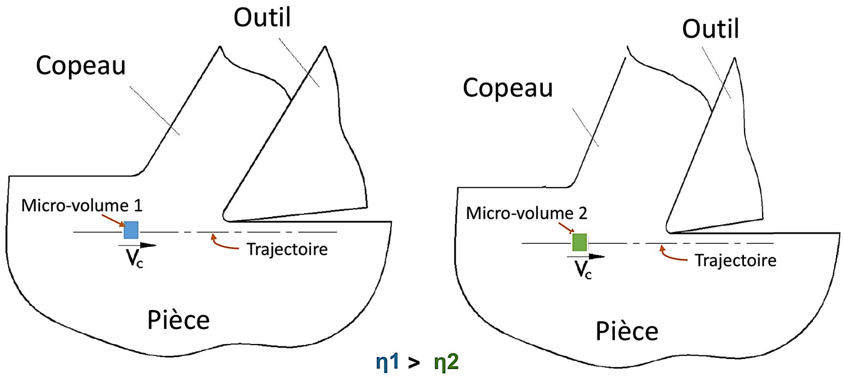
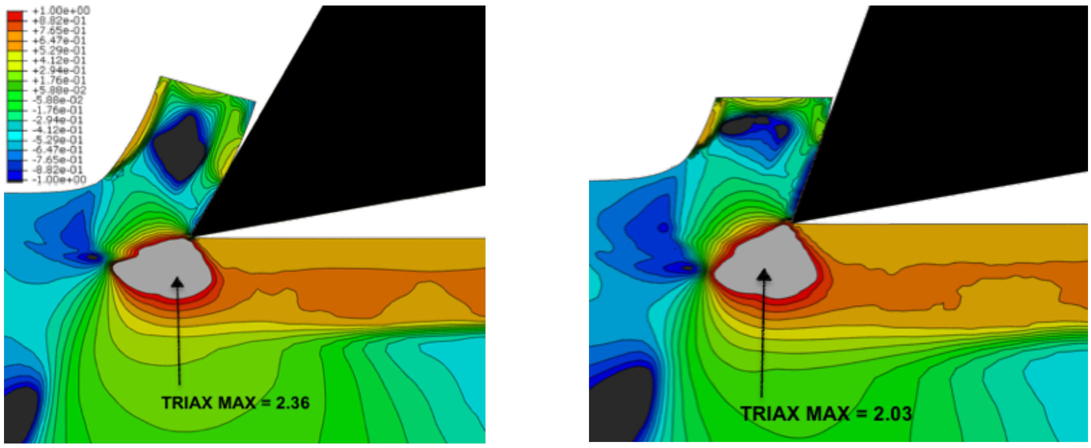


Figure 36. *Positions of the micro-volumes concerning the comparison of the tool's loading cycles on the surface creating different stresses triaxialities due to the variation of cutting geometry.*

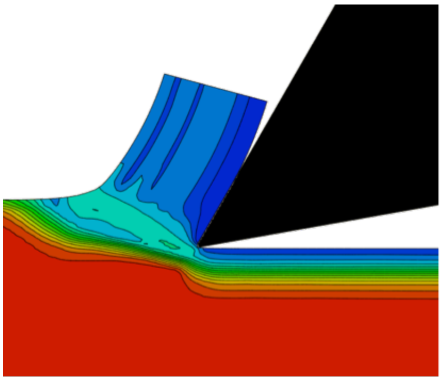


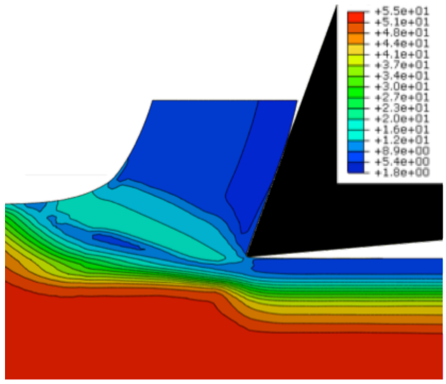
(a) *(b)*

Figure 37. *Chartes of simulated stresses triaxialities while orthogonal cutting for: (a) γ = 30°, α = 10°, h = 0.2 mm and V_c = 90 m/min ; (b) γ = 20°, α = 10°, h = 0.2 mm and V_c = 90 m/min.*

I.5.1.3 Relation between the micro-hardness, the grain size, and the cutting conditions

Micro-hardness and the grain size are associated to the interaction ($\gamma \times \alpha$), then ($h \times \alpha$), and finally with less relevance to the angles α and γ. As the hardness is related to the grains size, it seems normal that those parameters show analogue correlations with the cutting conditions. If this effect is considered, particularly in the grains size, it is well visible in the numerical simulations (cf. Figure 38). In fact, the tool geometry associated to the uncut chip thickness h affects the state of stress: when the rake angle decreases, stresses triaxiality decreases, then the equivalent plastic strain at damage initiation raises and consequently plastic strain increases. An increase in the plastic strain implies an increase in dislocations density, then in micro-hardness. Moreover, in absence of microstructural transformations such as dynamic recrystallization in the case of OFHC copper (because the recrystallization threshold is not reached), grains break or are elongated and deformed according to a priviledged direction (making an inclination angle regarding the cutting direction).



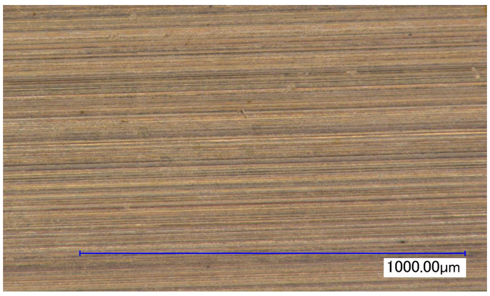


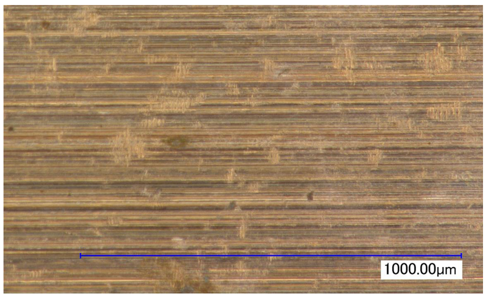
(a) *(b)*

Figure 38. *Repartition of the simulated grains size (in µm) at (a) $\gamma = 30°$, $\alpha = 10°$, $h = 0.2$ mm andt $V_c = 90$ m/min ; (b) $\gamma = 20°$, $\alpha = 5°$, $h = 0.2$ mm and $V_c = 90$ m/min.*

I.5.1.4 Relation between the roughness and the cutting conditions

The surface roughness at maximum height S_t is found in association with the cutting tool geometry, in particular with the interaction ($\gamma \times \alpha$). Characterising the maximum height profile and in absence of machining striae ruled by the tool geometry, it reflects the existence of micro-tearing and adhesion of the material on the generated surface. In fact, in the case of orthogonal cutting, roughness is related to the profile of the cutting edge (tool wear, but it is not taken into account in this study that all trials were performed with new tools), but it is essentially related to the tool edge geometry defaults and to the adhesive friction between the flank face and the surface of the workpiece, frequently seen in the case of ductile materials (it is the case of annealed copper, the workmaterial) when it is machined with small relative cutting edges inducing micro-tearing (Zorev 1966). It is exactly the case observed while machining copper (cf. Figure 39). They are due to a ratio between the shear stress and the normal stress inside the zone adjacent to the cutting edge radius at the contact tool/workpiece (flank contact) near 1, even higher than 1, creating conditions in favor of adhesion phenomenon [18]. So, a small rake angle raises the shear stress in this zone, associated to a small flank angle, they induce adhesion.





(a) *(b)*

Figure 39. *Micro-tearings in the case of orthogonal cutting of OFHC copper (a) $\gamma = 20°$, $\alpha = 10°$, $h = 0.05$ mm and $V_c = 120$ m/min ; (b) $\gamma = 20°$, $\alpha = 5°$, $h = 0.05$ mm and $V_c = 120$ m/min.*

I.5.2 Physical origins of the impact of surface integrity parameters on corrosion resistance

In this paragraph, the physical origin of the impact of each parameter previously presented on the surfaces electrochemical behavior and their corrosion resistance will be explained.

I.5.2.1 Impact of residual stresses and the plastic layer on corrosion resistance

It is fundamental that que the amplitude of the residual stresses affects the free energy of the material that, in return, has an impact on the work function of the electrons on the surface (Ralston and Birbilis 2010). the residual stresses σ_x then σ_y are shown as the first order parameters influencing the local electrochemical behavior and the corrosion resistance. In fact, resulting from the common action of corrosion and the tensile residual stress, this corrosion phenomenon appears in the surface. Being passive, its protecting oxide film is locally broken under the action of tensile stresses, creating vacancies in the film [20]. The conductivity of the passive film increases leading to a local corrosion. For that reason, increasing tensile residual stresses accelerates the anodic reaction (dissolution of the surface). However, in what concerns the plastic layer and the micro-hardness and their impact that appears only while long term aging, it can be explained as follows: first, plastic strain gradients induce residual stresses, and that send us back to the impact of stresses on corrosion resistance described previously. Then, after the dissolution of the smallest grains on the surface, the plastic layer presents bigger grains size, but in-depth elongated. So, intergranular corrosion is more susceptible to be developed in-depth when the plastic layer is thicker.

I.5.2.2 Impact of the microstructure and the micro-hardness on corrosion

The grains size has an important impact on the electrochemical behavior and on the corrosion resistance of the machined surface. In fact, it affects the energy of formation of the charged defaults during the thickening of the passive film and influences the exchanges inside the film of oxide. it has been demonstrated in literature that a high density of grains boundaries is in favor of the rapid formation of the passive film as the electrons are more active at the boundaries (Ralston and Birbilis 2010). Moreover, in some conditions, grains boundaries are the siege of highly important localized corrosion when the remaining material is intact (intergranular corrosion). On the other side, when the grain size is small, the grains boundaries number is high and intergranular corrosion is less deep regarding to the surface.

About the dispersions seen in the measurement of the pitting potential, they can be explained by the heterogeneity of the machined surface microstructure at the local scale. In fact, it is worth to notice that the measured grain size is a mean value, and a local electrochemical analysis performed on an area containing a majority of small grains won't give the same electrochemical response as an analysis performed on an area containing big grains with small ones, but resulting into the same mean grain size.

Concerning to the micro-hardness, as it is highly tied to the plastic strain and the grains size, it is natural that it appears as a factor highly correlated to the electrochemical behavior of the surfaces.

I.5.2.3 Impact of the topography on the corrosion resistance

According to the statistical analysis, the surface roughness at maximum height S_t is a parameter having a non negligible influence on the electrochemical reactivity of the surfaces. Many hypotheses can be proposed to explain this correlation. In fact, it is demonstrated that the roughness affects the electronic behavior, described by the electronic work function (*EWF*) of copper (Li and Li 2006). The mean value of the *EWF* decreases with the increase of the roughness, but the local fluctuation of the EWF increases. Such fluctuation is in favor of the formation of micro-electrodes and so, accelerates corrosion. On the other hand, the density of the local charges can raise with the increase of the local roughness. Such an heterogeneous distribution of the charges density promotes pitting initiation [21].

I.5.2.4 Chemistry of copper corrosion

reactions taking place while copper corrosion under saline atmosphere / saline solution of NaCl can be explained by the description given in the reference [16]. At corrosion potential, the anodic dissolution of copper can be represented by the reactions:

$$Cu \rightarrow Cu^{+} + e^{-}$$ **Eq. 4**

$$Cu^{+} + Cl^{-} \rightarrow CuCl$$ **Eq. 5**

For low chloride concentrations, the cations Cu^{2+} are produced according to the reaction (Eq. **6**) while for chloride concentrations higher than 0.1 M, copper anodic dissolution produces anions of $CuCl_2^-$ (Eq. **7**).

$$CuCl \rightarrow Cu^{2+} + Cl^{-} + e^{-}$$ **Eq. 6**

$$CuCl + Cl^{-} \rightarrow CuCl_2^{-}$$ **Eq. 7**

The production of atacamite from Cu^+ can follow different reactions: (Eq. **8** et Eq. **9**)

$$12CuCl_2^{-} + 6H_2O + 3O_2 \rightarrow 4Cu_2(OH)_2Cl + 4Cu^{2+} + 20Cl^{-}$$ **Eq. 8**

$$12CuCl + 6H_2O + 3O_2 \rightarrow 4Cu_2(OH)_2Cl + 4Cu^{2+} + 8Cl^- \qquad \textbf{Eq. 9}$$

If CuCl produces cations Cu^{2+} according to reaction (Eq. **6**), then the atacamite is produced according to the reaction (Eq. **10**). From this reaction, protons are released in the droplets, leading to a decrease of the pH. This promotes the depassivation and the general dissolution under the droplets. As the general dissolution is experimentally observed, the reaction (Eq. **10**) can be dominant, it means that cations Cu^{2+} are released in the solution during the dissolution of copper (Eq. **6**) instead of the anions $CuCl_2^-$.

$$2Cu^{2+} + 3H_2O + Cl^- \rightarrow Cu_2(OH)_3Cl + 3H^+ \qquad \textbf{Eq. 10}$$

The cathodic reactions correspond either to a reduction of the oxygen (Eq**. 11**) or to a reduction of the protons H^+. However, the cathodic reaction can be taken over by the reduction of oxygen. In fact, the evolution of hydrogen is significant only at more negative potentials [22].

$$O_2 + 2H_2O + 4e^- \rightarrow 4OH^- \qquad \textbf{Eq. 11}$$

Conclusion

A statistical multi-physical analysis using the correlations of Pearson and the analysis of variance (*2 ways ANOVA*) as techniques is applied to the results issued from the cutting operations, the surface integrity, the electrochemical analysis and the corrosion of the surfaces obtained by orthogonal cutting of OFHC copper. Physical explanations of the observed phenomena besides corrosion mechanisms are given.

The results expose a great influence of the surface integrity on the material dissolution and on the risk of corrosion. Then, it is proven that an increase of the tensile residual stresses is susceptible of the acceleration of the anodic reaction (dissolution of the surface) highlighting the risk of developing pitting corrosion. Moreover, the thickening of the plastic layer seems harmful for long term aging, besides a rough surface is more susceptible to develop dissolution. Nevertheless, grains refinement at the surface and the raise of hardness are in favor of the metal passivation and delay the risk of pitting. making the link with the cutting conditions, for a better corrosion resistance of OFHC copper surfaces obtained by finishing machining, it is recommended to maximize the thickness of cut h and the cutting speed V_c leading to a reduction of the tensile residual stresses susceptible of promoting the breakage of the passive film of oxide protecting the copper. then, it is recommended to avoid low flank angles α in order to minimize adhesive friction in the flank face leading to micro-tearing which represent favorable sites for dissolution initiation. However, concerning to the rake angle γ, in contrary to what is usually recommended for ductile materials machining in order to minimize cutting energy, the reduction of the rake angle can be beneficial for the sustainability of the machined surface. In fact, a lower rake angle leads to a higher surface hardness and smaller grains size what complicates the development of the dissolution process.

Definitely, a phenomenological analysis is developed in the perspective of giving physical explanations to the phenomena taking place and justifying the link between the cutting conditions, the surface integrity and the corrosion resistance.

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